



CHAPTER 4 & 5

Limits Systems and Tolerance

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October 2025

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Introduction

- Experience shows that it is impossible to make parts (manufacture) economically to the exact dimensions indicated on a drawing.

This may be due to,

- (i) Inaccuracies of machines and tools,
- (ii) Inaccuracies in setting the work to the tool, and
- (iii) Error in measurement, etc.

Therefore;

1. In order that assembled parts to function properly and
2. To allow for interchangeable manufacturing, it is necessary to permit only a certain amount of tolerance on each parts of the mating parts and a certain amount of allowance between them.
3. Moreover, the workman, has to be given some allowable margin (tolerance) so that he can produce a part, the dimensions of which will lie between two acceptable limits, a maximum and a minimum.

Terminologies

1. **Tolerance:** Tolerance is the permissible limit or limits of variation in a physical dimension.
 - It defines how much a dimension can **deviate** from its **nominal value**.
 - **Example:** For a shaft with a nominal diameter of **10 mm**, a tolerance might specify that the actual diameter can vary from **9.95 mm** to **10.05 mm**.
2. **Allowance:** is the **intentional difference** between the sizes of **mating parts**.
 - It represents the **minimum clearance** or **interference** that ensures proper fit between parts.
 - **Example:** If a hole is designed to fit a shaft, the allowance might specify that the hole is **0.1 mm** larger than the shaft's **maximum diameter**, ensuring a proper fit.

3. Fit: refers to the relationship between **two mating parts**, defined by the tolerances and allowances.

There are different types of fits, such as:

- **Clearance Fit:** Always provides space between the parts.
- **Interference Fit:** Always results in a tight fit, with parts overlapping.
- **Transition Fit:** Can result in either a clearance or interference fit, depending on the actual sizes.

4. Limit: is the **maximum** and **minimum size** of a dimension, defining the range in which the actual size of a part must fall. Limits are often specified in pairs, such as:

- The upper limit and,
- Lower limit.

• **Example:** For a hole with a limit of **10 mm $\pm$ 0.1 mm**, the upper limit would be **10.1 mm** and the lower limit would be **9.9 mm**.

For instance,

- ❖ A dimension $\Phi 15.02 \pm 0.004$ means the manufactured part can be anywhere between 15.024 and 15.016.
- ❖ The **tolerance permitted** (Max-Min)=**0.008**
- ❖ The **limits** are **15.024** and **15.016**.

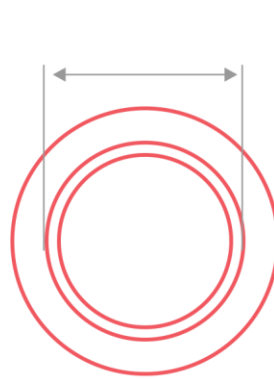
Limit system

- The system in which **a variation is accepted** is called the **limit system** and the **allowable deviations** are called **tolerances**.
- The relationships between the mating parts are called fits.
- “The study of limits, tolerances and fits are a must for technologists involved in production.”
- The same must be reflected on production drawing, for guiding the craftsman on the shop floor.
- The **two extreme permissible** sizes between which the **actual size is contained** are called **limits**.
- The **maximum size** is called the **upper limit** and the **minimum size** is called the **lower limit**.

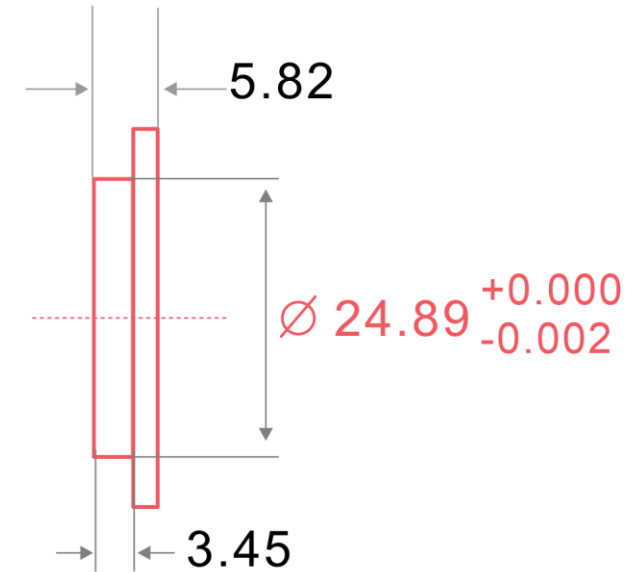
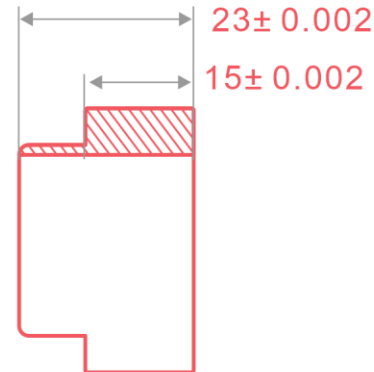
Tolerance

- Is the permissible variation of a size (it can be a hole, shaft, etc).
- The difference between the maximum and minimum permissible limits of the given size (shaft, hole size).
- If the variation is provided on **one side of the basic size**, it is termed as unilateral tolerance.
- On the other side, if the variation is provided on **both sides of the basic size**, it is known as bilateral tolerance.

Example



Bilateral Tolerance



Unilateral Tolerance

Example

Example

$$100^{+0.059}_{+0.037}$$

Given

— Bore Diameter(mm) = 100mm

— Upper limit = $100 + 0.059 = 100.059\text{mm}$

— Lower limit = $100 + 0.037 = 100.037\text{mm}$

Tolerance = 0.022mm

- Therefore, the **actual dimension** of the finished bore should lie between **100.059** and **100.037mm**.

Tolerance

```
graph TD; Tolerance[Tolerance] --> Dimensional[Dimensional]; Tolerance --> Geometrical[Geometrical];
```

Dimensional

Definition: Dimensional tolerance specifies the allowable variation in size for a given dimension.

- It defines how much a dimension can deviate from its nominal value.

Focus: Primarily concerned with linear measurements (length, width, height, diameter, etc.).

Example: A shaft may have a nominal diameter of 10 mm with a tolerance of ± 0.1 mm, allowing for a range of 9.9 mm to 10.1 mm.

Geometrical

Definition: Geometric tolerance defines the allowable variation in the shape, orientation, and location of a feature.

- It specifies the permissible deviation from the ideal geometric form.

Focus: Concerned with the geometry of features, including flatness, straightness, circularity, cylindricity, profile of a surface, angularity, perpendicularity, parallelism, and location (such as position and concentricity).

Example: A flat surface may have a flatness tolerance of 0.05 mm, meaning the surface must remain within this deviation from the ideal flat plane.

Deviation;

- **Deviation:** is the algebraic difference between **a size** (actual, maximum, minimum, etc.) and the corresponding **basic size**.
- **Actual deviation** = actual size - basic size.
- **Upper deviation** = maximum limit size - basic size.
- **Lower deviation** = minimum limit size - basic size.
- **Actual size:** obtained after manufacturing.

Example 1 (a)

- For the dimension value

$$100 \begin{matrix} +0.059 \\ +0.037 \end{matrix}$$

- The **upper deviation** $= +0.059 = (100.059 - 100)$
- The **lower deviation** $= +0.037 = (100.037 - 100)$
- **For example Actual deviation** $= (100.045 - 100) = +0.045$

Allowance

- It is the dimensional difference between the **maximum material limits(upper limits)** of the mating parts (hole-shaft mating).
- If the **allowance is positive**, it will result in minimum clearance between the **mating parts** and
- If the **allowance is negative**, it will result in maximum interference between **mating parts**.

Defining that;

- **A shaft** is any **external feature**(cylindrical or non-cylindrical features)
- **A hole** is any **internal features** (grooves, slots, etc)

Example 1 (b)

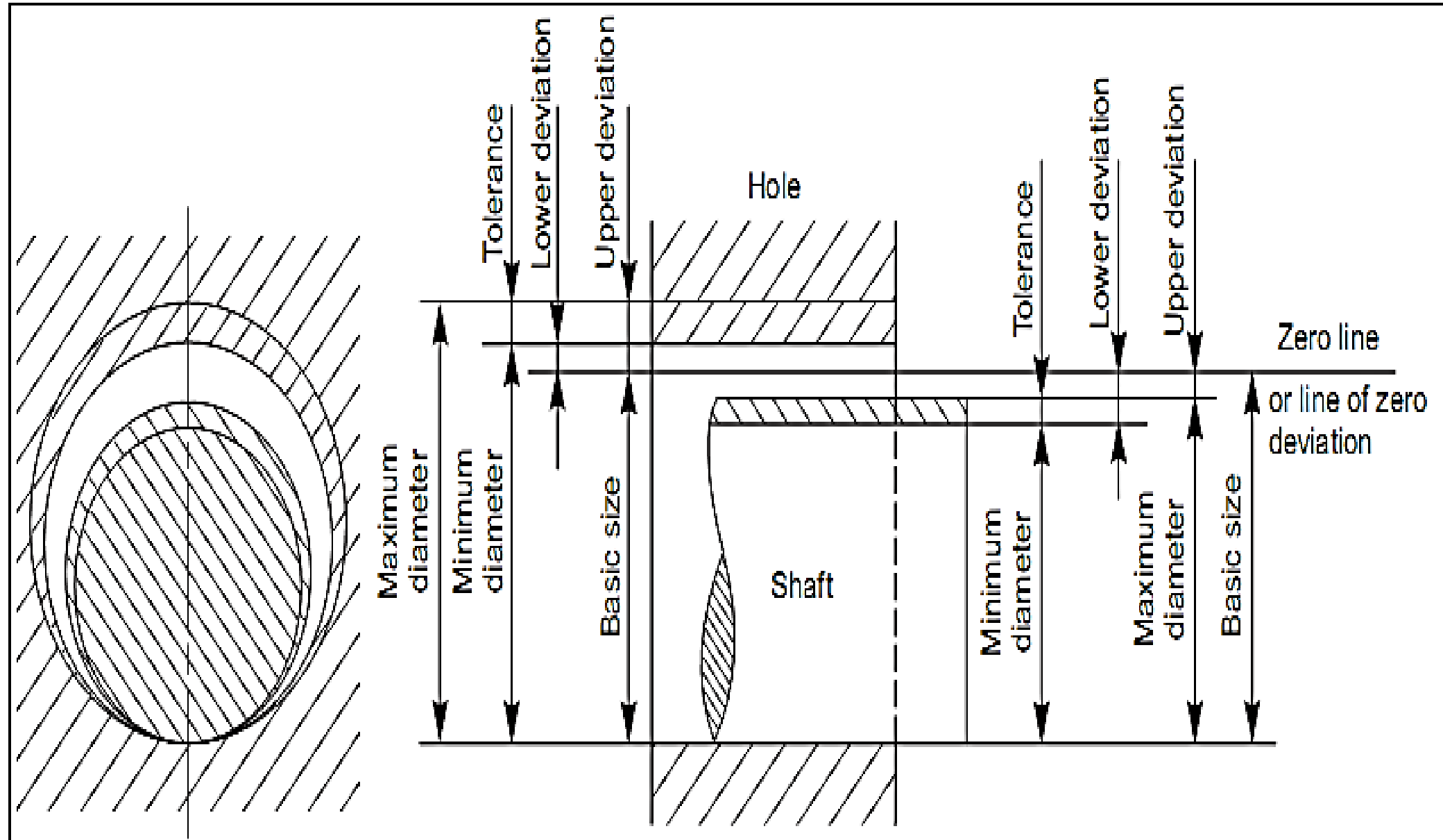
- Given
 - Shaft size = $49.90 \pm 0.02\text{mm}$
 - Hole size $50.0 \pm 0.05\text{mm}$

Allowance

- Maximum shaft size = $(49.90 + 0.02) = 49.92\text{mm}$
- Maximum hole size = $(50.0 + 0.05) = 50.05\text{mm}$
- Maximum hole size - maximum shaft size = 0.13mm

Allowance = 0.13mm.

Basic size, deviations and tolerances



Recommendations on Tolerances

- **Great care** and **judgment** must be exercised in deciding the tolerances which may be applied on various dimensions of a component.

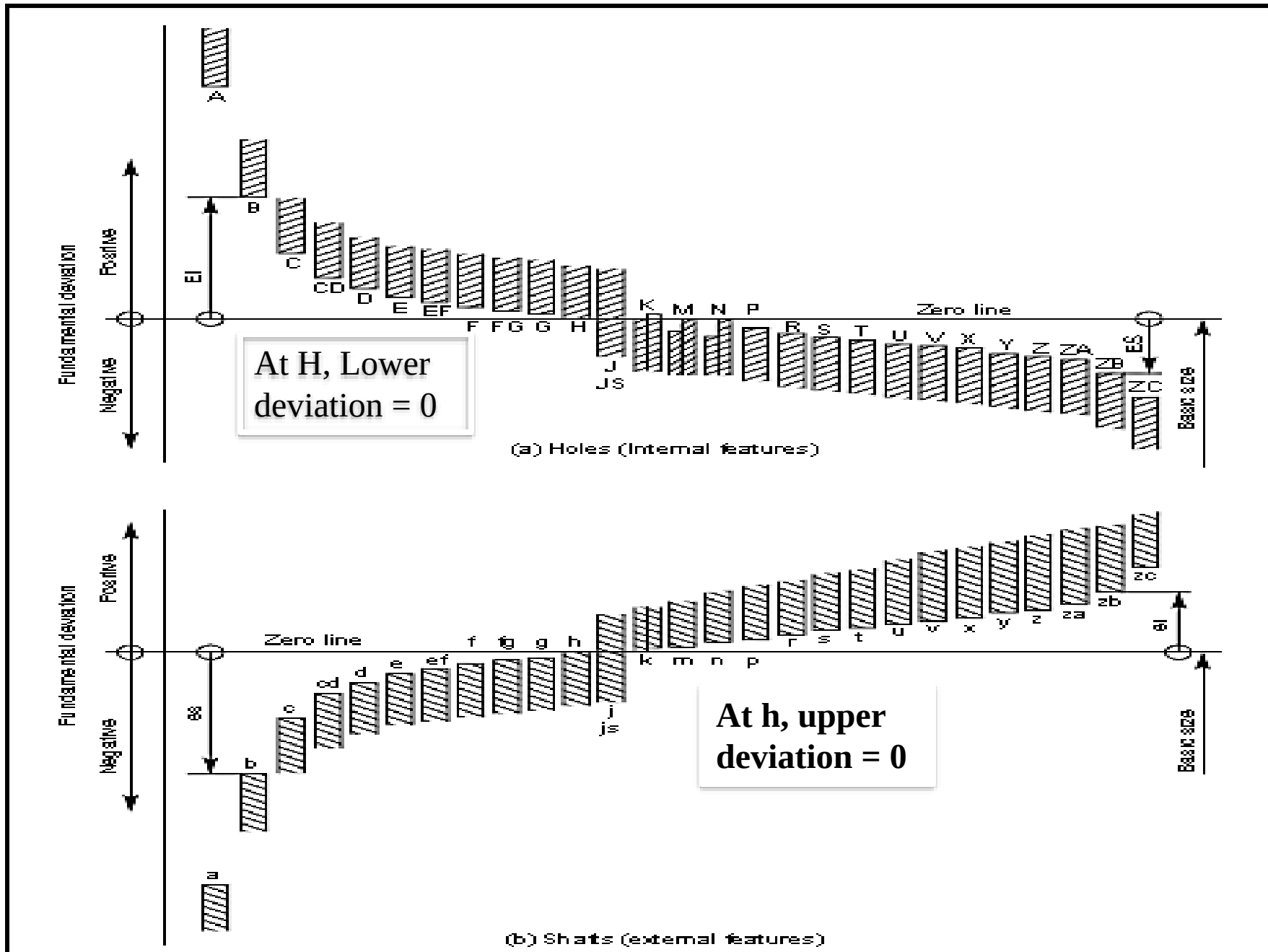
Physical interpretation

- If **tolerances are to be minimum**, that is, if the **accuracy requirements are severe**, the **cost of production increases**.
- Hence, “**maximum possible tolerances must be recommended**” wherever possible.

Fundamental Tolerance

- Tolerance is denoted by two symbols,
 - a **letter symbol** and
 - a **number symbol**, called the **grade**.
- The letter symbols range from
 - **A to ZC for holes(**capital letters**)** and
 - **a to zc for shafts(**small letters**)**.
- These letter represent **the degree of closeness of the tolerance zone** (positive or negative) to the basic size.

Tolerance zones



Empirical formulae

- The fundamental tolerance is given by

$$i = 0.45 \times \sqrt[3]{D} + 0.001 D$$

Where i-standard tolerance unit(microns)

D-the geometrical mean of the limiting values(mm)

Grade and Tolerance Values

Table 15.1A Relative magnitude of IT tolerances for grades 5 to 16 in terms of tolerance unit i for sizes upto 500 mm

Grade	IT 5	IT 6	IT 7	IT 8	IT 9	IT 10	IT 11	IT 12	IT 13	IT 14	IT 15	IT 16
Tolerance values	$7i$	$10i$	$16i$	$25i$	$40i$	$64i$	$100i$	$160i$	$250i$	$400i$	$640i$	$1000i$

Table 15.1 Fundamental tolerances of grades 01, 0 and 1 to 16 (values of tolerances in microns) (1 micron = 0.001 mm)

Diameter steps in mm	Tolerance Grades																	
	01	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14*	15*	16*
To and inc 3	0.3	0.5	0.8	1.2	2	3	4	6	10	14	25	40	60	100	140	250	400	600
Over 3																		
To and inc 6	0.4	0.6	1	1.5	2.5	4	5	8	12	18	30	48	75	120	180	300	480	750
Over 6																		
To and inc 10	0.4	0.6	1	1.5	2.5	4	6	9	15	22	36	58	90	150	220	360	580	900
Over 10																		
To and inc 18	0.5	0.8	1.2	2	3	5	8	11	18	27	43	70	110	180	270	430	700	1100
Over 18																		
To and inc 30	0.6	1	1.5	2.5	4	6	9	13	21	33	52	84	130	210	330	520	840	1300
Over 30																		
To and inc 50	0.6	1	1.5	2.5	4	7	11	16	25	39	62	100	160	250	390	620	1000	1600
Over 50																		
To and inc 80	0.8	1.2	2	3	5	8	13	19	30	46	74	120	190	300	460	740	1200	1900
Over 80																		
To and inc 120	1	1.5	2.5	4	6	10	15	22	35	54	87	140	220	350	540	870	1400	2200
Over 120																		
To and inc 180	1.2	2	3.5	5	8	12	18	25	40	63	100	160	250	400	630	1000	1600	2500
Over 180																		
To and inc 250	2	3	4.5	7	10	14	20	29	46	72	115	185	290	460	720	1150	1850	2900
Over 250																		
To and inc 315	2.5	4	6	8	12	16	23	32	52	81	130	210	320	520	810	1300	2100	3200
Over 315																		
To and inc 400	3	5	7	9	13	18	25	36	57	89	140	230	360	570	890	1400	2300	3600
Over 400																		
To and inc 500	4	6	8	10	15	20	27	40	63	97	155	250	400	630	970	1550	2500	4000

*Upto 1 mm, Grades 14 to 16 are not provided.

Fundamental Deviation

Table 15.2 Fundamental deviations for shafts of types **a** to **k** of sizes upto 500mm (*contd.*)

Fundamental deviation in microns										(1 micron = 0.001 mm)					
Diameter steps in mm		Upper deviation (<i>es</i>)								js*	Lower deviation (<i>ei</i>)				
		<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>		<i>j</i>			<i>k</i>	
over	upto	All grades									5.6	7	8	4 to 7	≤ 3, > 7
—	*3	− 270	− 140	− 60	− 20	− 14	− 6	− 2	0	± IT/2	− 2	− 4	− 6	− 0	− 0
3	6	− 270	− 140	− 70	− 30	− 20	− 10	− 4	0		− 2	− 4	—	+ 1	0
6	10	− 280	− 150	− 80	− 40	− 25	− 13	− 5	0		− 2	− 5	—	+ 1	0
10	14	− 290	− 150	− 95	− 50	− 32	− 16	− 6	0		− 3	− 6	—	+ 1	0
14	18														
18	24	− 300	− 160	− 110	− 65	− 40	− 20	− 7	0		− 4	− 8	—	+ 2	0
24	30														
30	40	− 310	− 170	− 120	− 80	− 50	− 25	− 9	0		− 5	− 10	—	+ 2	0
40	50	− 320	− 180	− 130											
50	65	− 340	− 190	− 140	− 100	− 60	− 30	− 10	0		− 7	− 12	—	+ 2	0
65	80	− 360	− 200	− 150											
80	100	− 380	− 220	− 170	− 120	− 72	− 36	− 12	0		− 9	− 15	—	+ 3	0
100	120	− 410	− 240	− 180											
120	140	− 460	− 260	− 200	− 145	− 85	− 43	− 14	0						
140	160	− 520	− 280	− 210							− 11	− 18	—	+ 3	0
160	180	− 580	− 310	− 230											

Table 15.2 Fundamental deviations for shafts of types **a** to **k** of sizes upto 500mm (*contd.*)

Fundamental deviation in microns										(1 micron = 0.001 mm)					
Diameter steps in mm		Upper deviation (es)								js+	Lower deviation (ei)				
		a	b	c	d	e	f	g	h		j			k	
over	upto	All grades								± IT/2	5.6	7	8	4 to 7	≤ 3, > 7
180	200	− 660	− 340	− 240	− 170	− 100	− 50	− 15	0						
200	225	− 740	− 380	− 260							− 13	− 21	—	+ 4	0
225	250	− 820	− 420	− 280											
250	280	− 920	− 480	− 300	− 190	− 110	− 56	− 17	0		− 16	− 26	—	+4	0
280	315	− 1050	− 540	− 330											
315	355	− 1200	− 600	− 360	− 210	− 125	− 62	− 18	0		− 18	− 28	—	+ 4	0
355	400	− 1350	− 680	− 400											
400	450	− 1500	− 760	− 440	− 230	− 135	− 68	− 20	0		− 20	− 32	—	+ 5	0
450	500	− 1650	− 840	− 480											

*The deviations of shafts of types **a** and **b** are not provided for diameters upto 1 mm

+ For types **js** in the particular Grades 7 to 11, the two symmetrical deviations $\pm IT/2$ may possibly be rounded, if the IT value in microns is an odd value; by replacing it by the even value immediately below.

Example 1 (c)

Calculate the fundamental tolerance for a shaft of 100 mm and grade 7.

The shaft size, 100 lies in the basic step, 80 to 120 mm and the geometrical mean is substituting,

$$D = \sqrt{80 \times 120} = 98 \text{ mm}$$

The tolerance unit, $i = 0.45 \sqrt[3]{98} + 0.001 \times 98 = 2.172$ microns

For grade 7, as per the Table 15.1A, the value of tolerance is,

$$16i = 16 \times 2.172 = 35 \text{ microns}$$

Fundamental deviations

- The symbols used in the Fig. for the fundamental deviations for the shaft and hole are as follows :

	Hole	Shaft
Upper deviation (E' cart superior)	ES	es
Lower deviation (E' cart inferior)	EI	ei
	es/ES-upper deviation	
	ei/EI-lower deviation	

Shafts, $ei = es - IT$

Holes, $EI = ES - IT$

Where **IT** is **fundamental tolerance of grade** obtained from Table 15.1.

Table 15.3 Fundamental deviations for holes of types A to N for sizes up to 500mm

Fundamental deviation in microns															(1 micron = 0.001 mm)					
Diameter steps in mm		Lower deviations (EI)										Upper deviations (ES)								
		A*	*B	C	D	E	F	G	H	Js+	J			K		M		N		
Over	Upto	All grades									6	7	8	≤ 8	> 8	≤ 8 ‡	> 8	≤ 8	> 8*	≤ 7
—	3*	+ 270	+ 140	+ 60	+ 20	+ 14	+ 6	+ 2	0	± IT/2	+ 2	+ 4	+ 6	0	0	− 2	− 2	− 4	− 4	Same deviation as for grades > 7 + Δ
3	6	+ 270	+ 140	+ 70	+ 30	+ 20	+ 10	+ 4	0		+ 5	+ 6	+ 10	− 1 + Δ	—	− 4 + Δ	− 4+Δ	− 8 + Δ	0	
6	10	+ 280	+ 150	+ 80	+ 40	+ 25	+ 13	+ 5	0		+ 5	+ 8	+ 12	− 1 + Δ	—	− 6+ Δ	− 6+Δ	− 10 + Δ	0	
10	14	+ 290	+ 150	+ 95	+ 50	+ 32	+ 16	+ 6	0		+ 6	+ 10	+ 15	− 1+ Δ	—	− 7 + Δ	− 7	− 12 + Δ	0	
14	18																			
18	24	+ 300	+ 160	+ 110	+ 65	+ 40	+ 20	+ 7	0		+ 8	+ 12	+ 20	− 2 + Δ	—	− 8 + Δ	− 8	− 15 + Δ	0	
24	30																			
30	40	+ 310	+ 170	+ 120	+ 80	+ 50	+ 25	+ 9	0		+ 10	+ 14	+ 24	− 2 + Δ	—	− 9 + Δ	− 9	− 17 + Δ	0	
40	50	+ 320	+ 180	+ 130																
50	65	+ 340	+ 190	+ 140	+ 100	+ 60	+ 30	+ 10	0		+ 13	+ 18	+ 28	− 2 + Δ	—	− 11+ Δ	− 11	− 20 + Δ	0	
65	80	+ 360	+ 200	+ 150																
80	100	+ 380	+ 220	+ 170	+ 120	+ 72	+ 36	+ 12	0	+ 16	+ 22	+ 34	− 3 + Δ	—	− 13 + Δ	− 13	− 23 + Δ	0		
100	120	+ 410	+ 240	+ 180																

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Table 15.3 Fundamental deviations for holes of types **A** to **N** for sizes upto 500mm (contd.)

A to N

Fundamental deviation in microns																	(1 micron = 0.001 mm)				
Diameter steps in mm		Lower deviations (EI)												Upper deviations (ES)							
		A*	*B	C	D	E	F	G	H	Js+	J			K		M		N			
Over	Upto	All grades									6	7	8	≤ 8	> 8	≤ 8 ‡	> 8	≤ 8	> 8*	≤ 7	
120	140	+ 460	+ 260	+ 200	+ 145	+ 85	+ 43	+ 14	0	± IT/2	+ 18	+ 26	+ 41	− 3 + Δ	—	− 15 + Δ	− 15	− 27 + Δ	0	Same deviation as for grades > 7 + Δ	
140	160	+ 520	+ 280	+ 210																	
160	180	+ 580	+ 310	+ 230																	
180	200	+ 660	+ 340	+ 240	+ 170	+ 100	+ 50	+ 15	0		+ 22	+ 30	+ 47	− 4+ Δ	—	− 17 + Δ	− 17	− 31 + Δ	0		
200	225	+ 740	+ 380	+ 260																	
225	250	+ 820	+ 420	+ 280																	
250	280	+ 920	+ 480	+ 300	+ 190	+ 110	+ 56	+ 17	0		+ 25	+ 36	+ 55	− 4+ Δ	—	− 20 + Δ	− 20	− 34 + Δ	0		
280	315	+ 1050	+ 540	+ 330																	
315	355	+ 1200	+ 600	+ 360	+ 210	+ 125	+ 62	+ 18	0		+ 29	+ 39	+ 60	− 4+ Δ	—	− 21 + Δ	− 21	− 37 + Δ	0		
355	400	+ 1350	+ 680	+ 400																	
400	450	+ 1500	+ 760	+ 440	+ 230	+ 135	+ 68	+ 20	0		+ 33	+ 43	+ 66	− 5+ Δ	—	− 23 + Δ	− 23	− 40 + Δ	0		
450	500	+ 1650	+ 840	+ 480																	

* The deviation of holes of types A and B in all grades >8 are not for diameters upto 1 mm.

+ For the hole of type Js in the grades 7 and 11, the two symmetrical ± deviations IT/2 may possibly rounded. If the IT value in microns is an odd value, replace it by the even value immediately below.

‡ Special case: For the hole M6, ES = 9 from 250 to 315 (instead of − 11).

Formulae for fundamental deviation for shafts up to 500 mm

Table 15.4

Upper deviation (<i>es</i>)		Lower deviation (<i>ei</i>)	
Shaft designation	In microns (for <i>D</i> in mm)	Shaft designation	In microns (For <i>D</i> in mm)
a	$= -(265 + 1.3D)$ for $D \leq 120$	k4 to k7	$= 0.6 \sqrt[3]{D}$
	$= -3.5 D$ for $D > 120$	k for grades ≤ 3 and ≥ 8	$= 0$
b	$\approx -(140 + 0.85 D)$ for $D \leq 160$	m	$= + (IT\ 7 - IT\ 6)$
	$\approx -1.8 D$ for $D > 160$	n	$= + 5 D^{0.34}$
c	$= -52 D^{0.2}$ for $D \leq 40$	p	$= + IT\ 7 + 0$ to 5
	$= -(95 + 0.8 D)$ for $D > 40$	r	$=$ geometric mean of values <i>ei</i> for <i>p</i> and <i>s</i>
d	$= -16 D^{0.44}$	s	$= + IT\ 8 + 1$ to 4 for $D \leq 50$
e	$= -11 D^{0.41}$		$= + IT\ 7 + 0.4 D$ for $D > 50$
f	$= -5.5 D^{0.41}$	t	$= IT\ 7 + 0.63 D$
g	$= -2.5 D^{0.34}$	u	$= + IT\ 7 + D$
h	$= 0$	v	$= + IT\ 7 + 1.25 D$
		x	$= + IT\ 7 + 1.6 D$
		y	$= + IT\ 7 + 2 D$
		z	$= + IT\ 7 + 2.5 D$
		za	$= + IT\ 8 + 3.15 D$
j5 to j8	no formula	zb	$= + IT\ 9 + 4 D$
		zc	$= + IT\ 10 + 5 D$ ²⁸

Example 2 Calculate the fundamental deviations for the shaft sizes given below :

(a) 30 e8 (b) 50 g6 (c) 40 m6.

From Table 15.4, the deviations for shafts are obtained :

(a) The upper deviation es for the shaft e

$$= -11 D^{0.41}$$

The value for $D = \sqrt{18 \times 30} = 23.24$ mm.

Hence, $es = -40$ microns (tallies with the value in Table 15.2).

(b) The upper deviation es for the shaft g

$$= -2.5 D^{0.34}$$

The value for $D = \sqrt{30 \times 50} = 38.73$ mm.

Hence, $es = -9$ microns (tallies with the value in Table 15.2)

(c) The lower deviation ei for the shaft m

$$= + (IT 7 - IT 6)$$

From the Table 15.1, the size 40 is in the range 30 and 50 and hence the mean diameter D , is 38.73 mm

$$\begin{aligned} \text{Tolerance unit } i &= 0.45 \sqrt[3]{D} + 0.001 D \\ &= 1.58 \text{ microns} \end{aligned}$$

The fundamental tolerance for grade 7, from the Table 15.1 is $16i$, i.e., 25 microns.

The fundamental tolerance for grade 6 is $10i$ or 16 microns.

Hence, $ei = 25 (IT 7) - 16 (IT 6) = +9$ microns (tallies with the value in Table 15.2).

Limit Dimensioning Techniques

3 Methods of placing limit dimensions

Method 1

- Here, the tolerance dimension is given by its basic value, followed by a symbol, comprising of both a letter and a numeral.
- The terms ϕ 25H7, 10H10 and 40C11 refer **to internal features**, since the terms involve **capital letter** symbols.

$$\phi 25H7 = \phi 25 \begin{matrix} +0.021 \\ +0.000 \end{matrix}$$

$$10H10 = 10 \begin{matrix} +0.058 \\ +0.000 \end{matrix}$$

$$40C11 = 40 \begin{matrix} +0.280 \\ +0.120 \end{matrix}$$

$$10h9 = 10 \begin{matrix} -0.000 \\ -0.036 \end{matrix}$$

$$\phi 25h9 = \phi 25 \begin{matrix} -0.000 \\ -0.052 \end{matrix}$$

$$\phi 40h11 = \phi 40 \begin{matrix} -0.000 \\ -0.160 \end{matrix}$$



Example $\phi 25 H7$

- The capital letter ' H ' $\Rightarrow$ *the lower deviation is zero*
- *The number symbol 7 $\Rightarrow$ the grade, the value of which is 21 microns (Table 15.1) which in turn is equal to the upper deviation.*

Example 40C11

- ❖ The number 40 $\Rightarrow$ the basic size
- ❖ The capital letter C $\Rightarrow$ *the lower deviations is 120 microns (Table 15.3).*
- ❖ The number 11 $\Rightarrow$ tolerance of grade 11(11i)
- ❖ $IT = 160 \text{ micron} = 0.160$
- ❖ Upper deviation(ES) = EI + IT
 $= 0.120 + 0.160 = 0.280$

External features

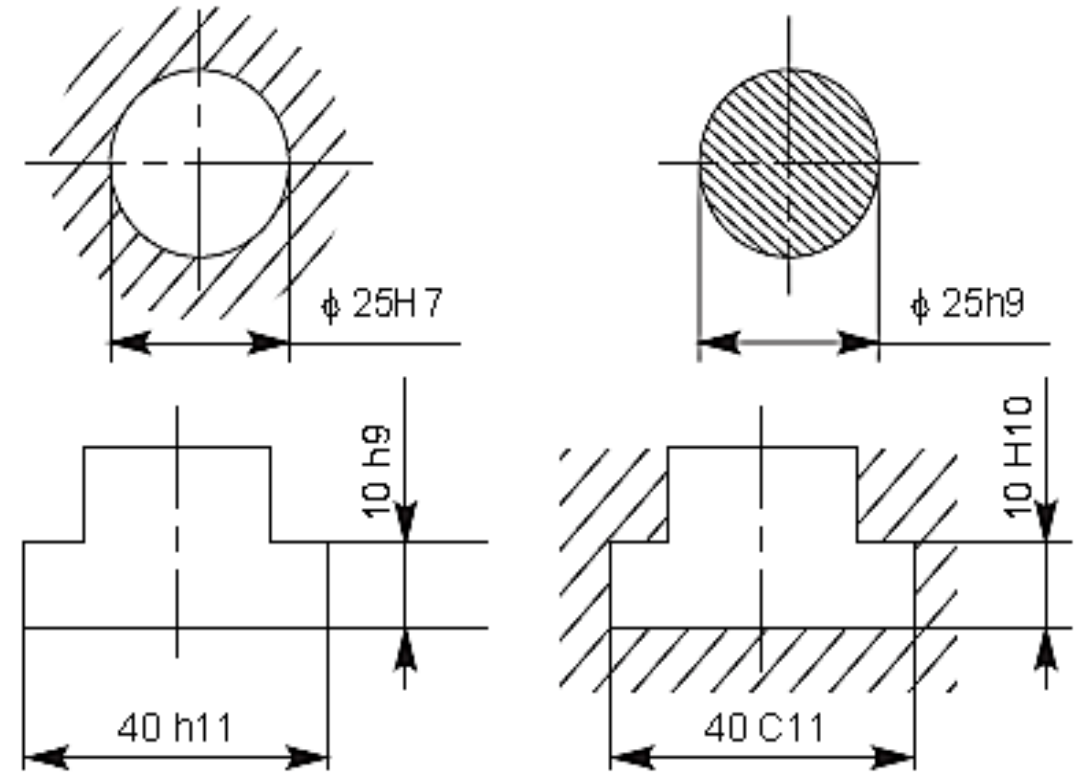
- The terms $\phi 40h11$ and $10h9$ refer to **external features**, since the terms involve **lower case** letters.
- The letter 'h' signifies that the upper deviation is zero.

- The number symbol 11 signifies the grade, the value of which is 160 microns (Table 15.1), which in turn is equal to the lower deviation.

$$160\text{micro} = 11i,$$

$$i = 0.45 * \text{cubrt}(D) + 0.001D$$

$$D = \text{sqrt}(30*50) = 38.73$$



Toleranced dimensions for internal and external features

Method-2

- Here, the basic size and the tolerance values are indicated above the dimension line;
- The tolerance values being in a size smaller than that of the basic size and the lower deviation value being indicated in line with the basic size.

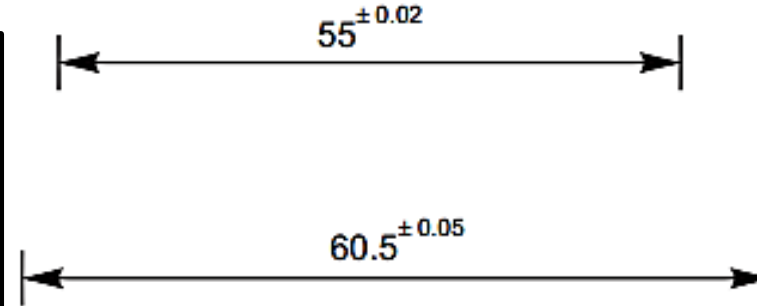


Fig. 15.5 Bilateral tolerance of equal variation

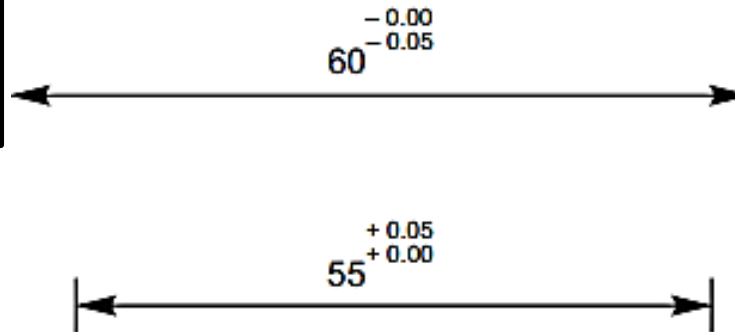


Fig. 15.7 Unilateral tolerance with zero variation in one direction

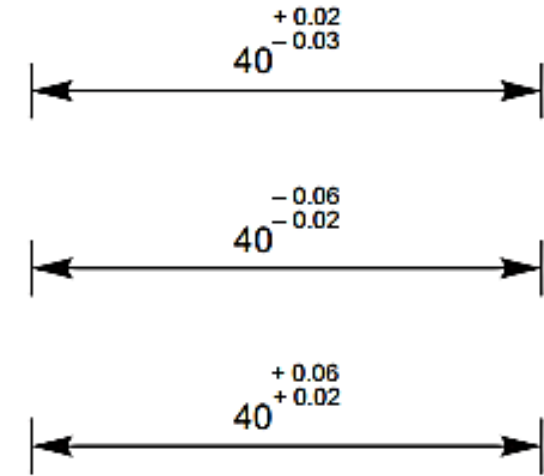


Fig. 15.6 Bilateral tolerance of unequal variation

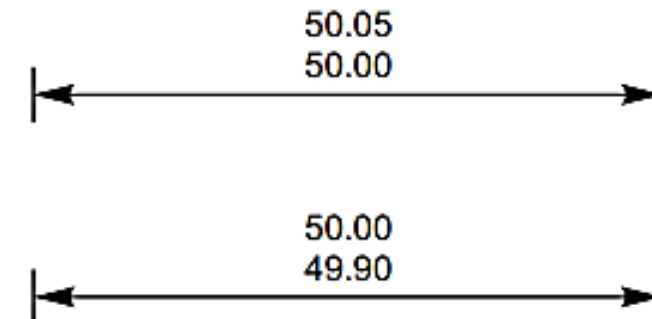


Fig. 15.8 Maximum and minimum size directly indicated

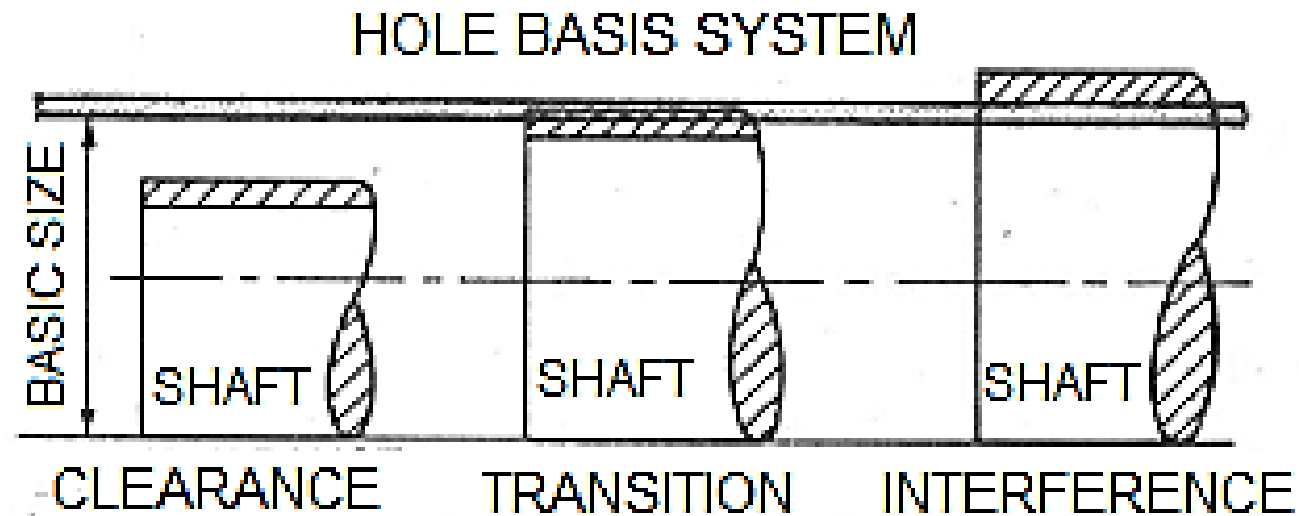
Method-3

- The maximum and minimum sizes are directly indicated above the dimension line

Basic Systems

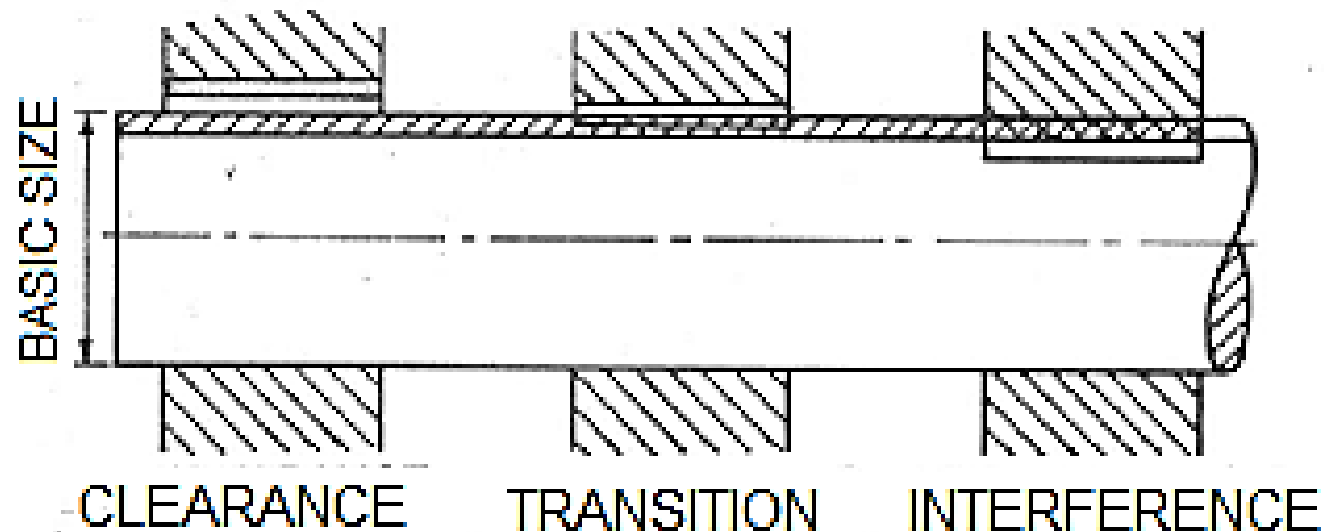
- ✓ *Hole Basic System*
- ✓ *Shaft Basic System*

Hole & Shaft Basis System



HOLE BASED SYSTEM

Size of the Hole is kept constant, Shaft size is varied to get different fits

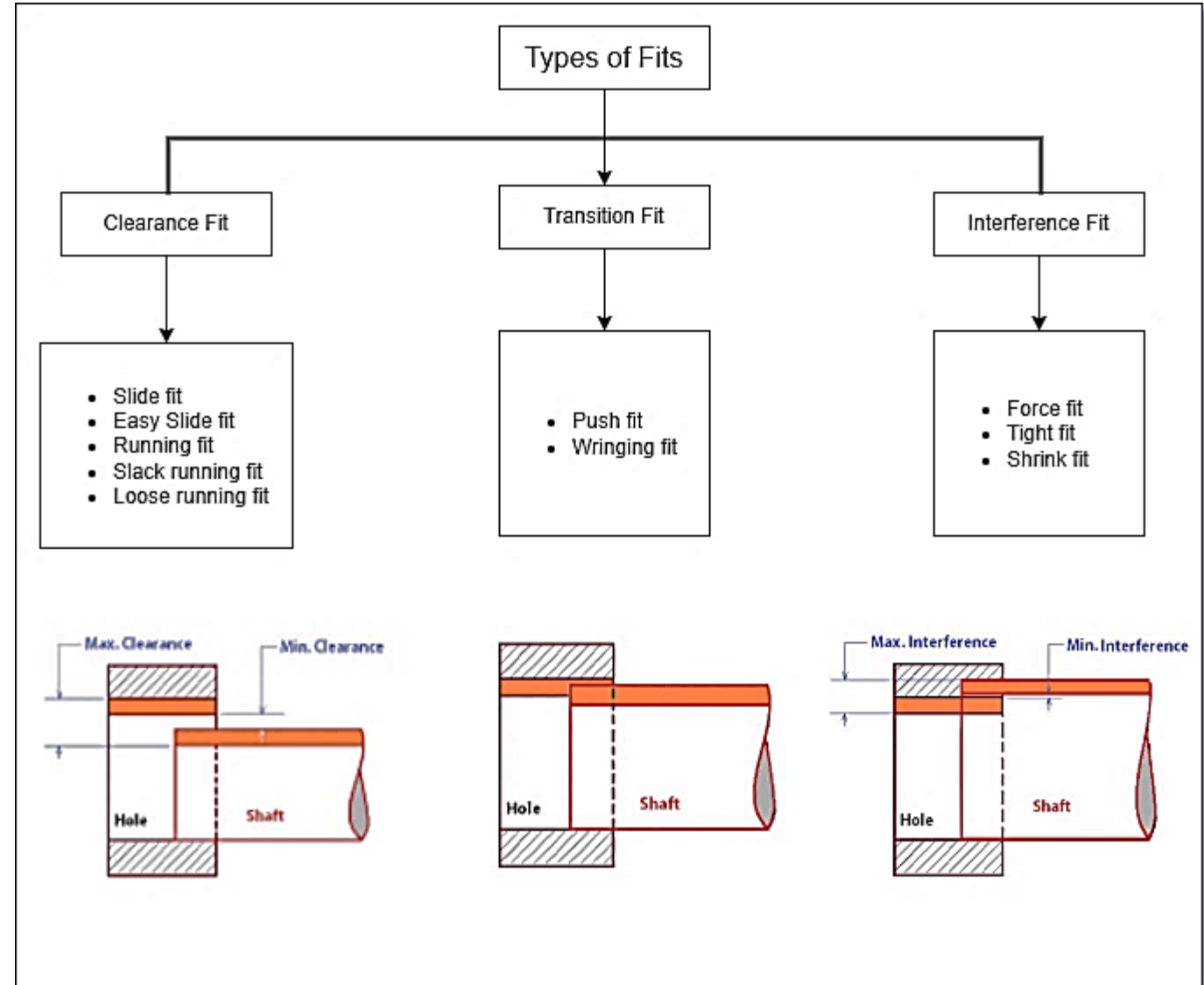


SHAFT BASED SYSTEM

Size of the Shaft is kept constant, Hole size is varied to get different fits

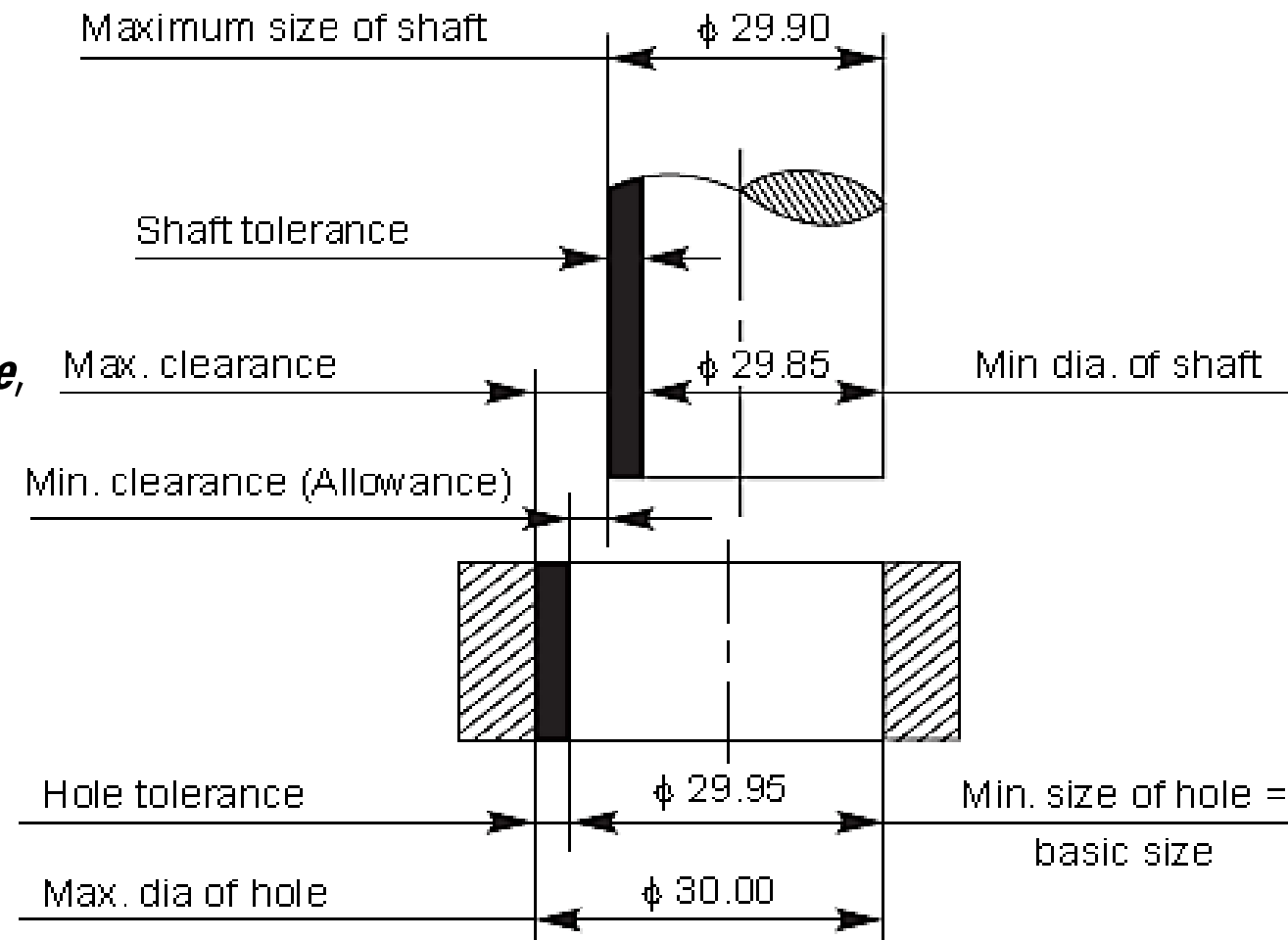
Fits

- The relation between two mating parts is known as a **fit**.
- Depending upon the **actual limits of the hole or shaft sizes**, fits may be classified as
 1. Clearance fit,
 2. Transition fit, and
 3. Interference fit.



Clearance fit

- The largest permissible diameter of the shaft is smaller than the diameter of the smallest hole.
- A clearance fit has **positive allowance**.
- **Allows rotation** or **sliding** between the mating parts.
- Used in:
 - **Tailstock spindle** of lathe machine,
 - Feed movement of the spindle quill in a **drill machine**,
 - **Cylinder and piston**,
 - **Shaft pulley**, etc.
- The clearance fits may be:
 - Sliding fit,
 - Easy sliding fit,
 - Running fit,
 - Slack running fit, and
 - Loose running fit.



FIGURE; CLEARANCE FIT

Types of clearance fit

Loose Fit

- It is used between those mating parts where **no precision** is required.
- It provides minimum allowance and is used on **loose pulleys, agricultural machineries** etc.

Running Fit

- For a running fit, the dimension of shaft should be **smaller enough** to maintain a **film of oil for lubrication**.
- It is used in **bearing** pair etc.,

Slide Fit/ Medium Fit

- It is used on those mating parts where **great precision** is required.
- It provides medium allowance and is used in tool **slides, slide valve, automobile parts**, etc.

For instance,

✓ In the above figure;

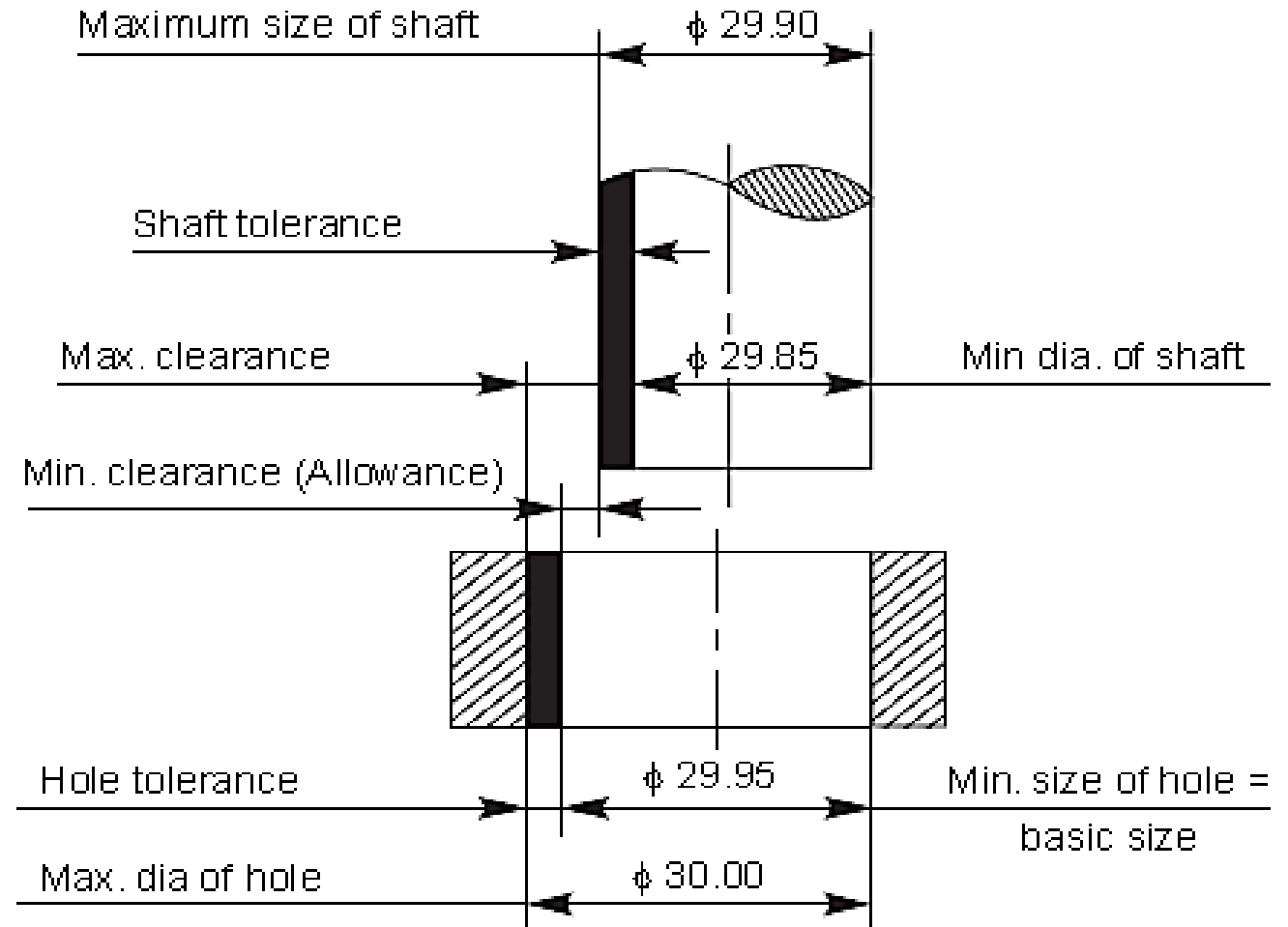
– The minimum clearance = (hole min size-shaft max size)

$$= 29.95 - 29.90 = 0.05$$

– The maximum clearance = (hole max size-shaft min size)

$$= 30.00 - 29.85 = 0.15$$

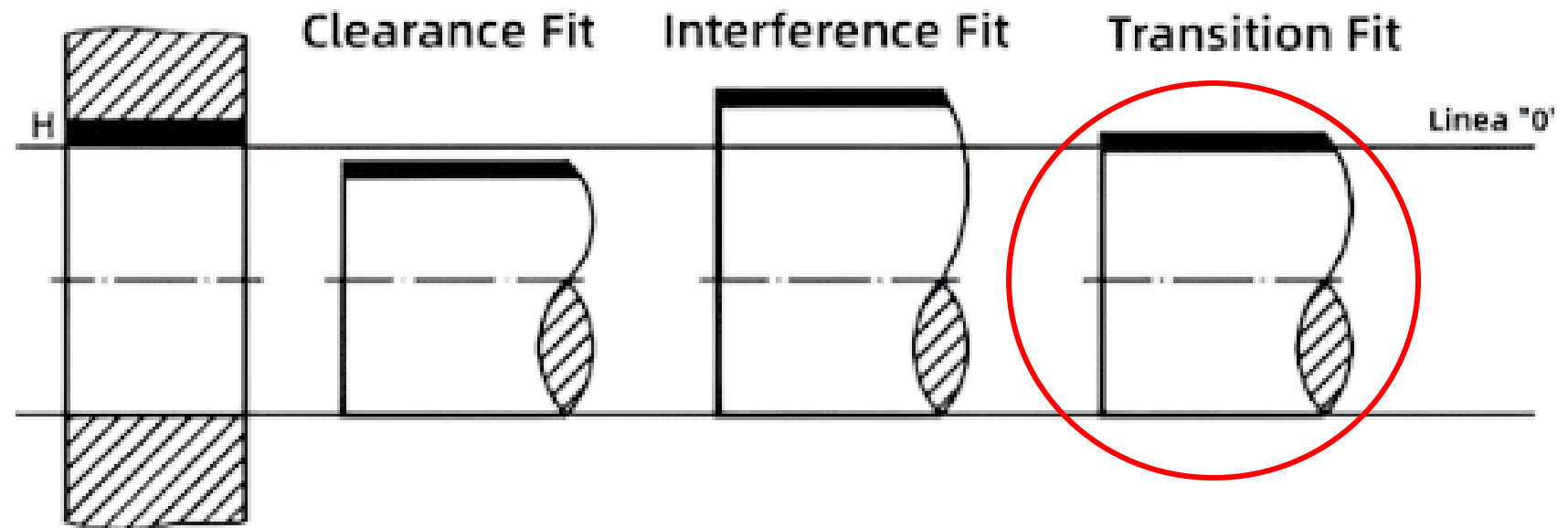
✓ We can see that both **minimum** clearance and **maximum** clearance are **positive** that confirms a clearance fit.



FIGURE; CLEARANCE FIT

Transition fit

- It may result in either clearance fit or interference fit depending on the **actual value** of the *individual tolerance* of the mating components.
- Transition fits are a compromise between *clearance* and *interference* fits.
- They are used for applications where accurate location is important but either a small amount of clearance or interference is permissible.
- Used in
 - *Railway wheels,*
 - *Fixing keys,*
 - *Changing gears,* etc.
- The transition fits may be:
 - Forcing fit,
 - Tight fit, and
 - Push fit.



Types of Transition fit

Push/ Snug Fit

- It refers to **zero allowance** and a **light pressure** is required in assembling the hole and the shaft.
- The moving parts show **least vibration** with this type of fit.

Force/ Shrink Fit

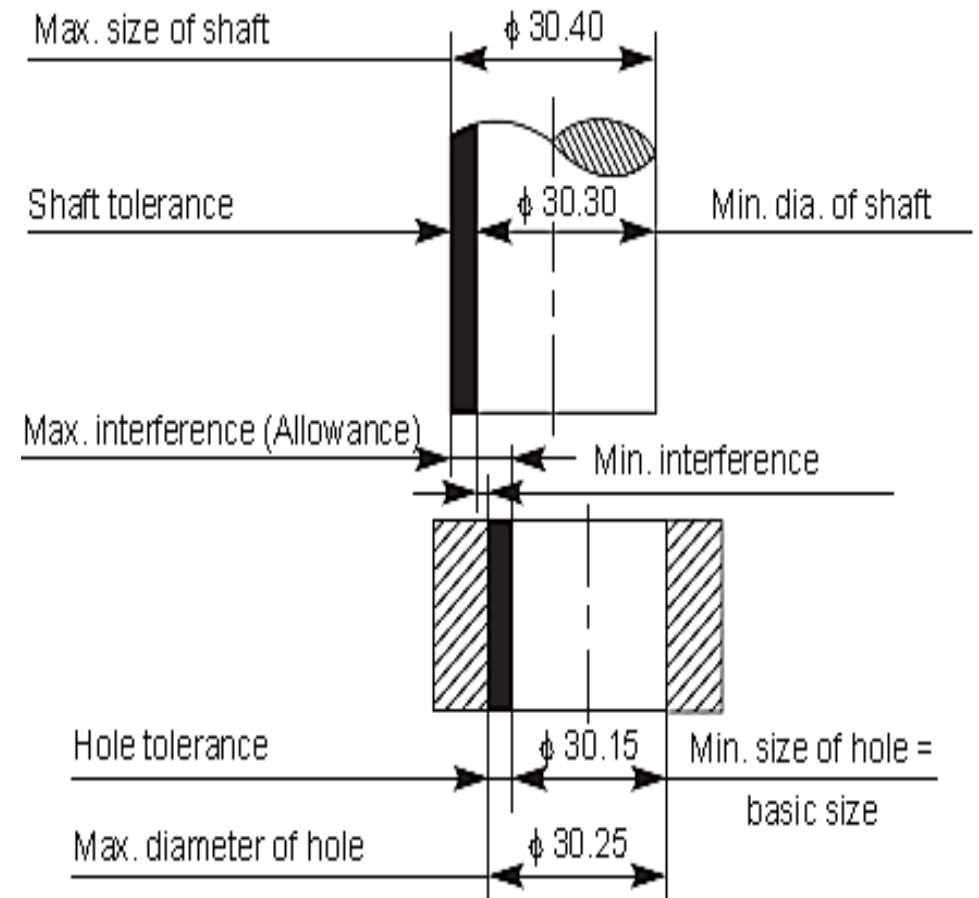
- A **force fit** is used when the two mating parts are to be rigidly fixed so that **one cannot move without the other**.
- It either requires **high pressure** to force the shaft into the hole or the hole to be **expanded by heating**.
- It is used in **railway wheels**, etc.,

Wringing Fit

- A **slight negative** allowance exists between two mating parts in wringing fit.
- It requires pressure to force the shaft into the hole and gives a light assembly.
- It is used in **fixing keys, pins**, etc.

Interference fit

- The minimum permissible diameter of the *shaft exceeds* the maximum allowable *diameter of the hole*.
- Interference fit has *negative allowance*.
- When two mating parts are assembled with an interference fit, it will be an *almost permanent assembly*, that is, the parts will not come apart or move during use.
- To assemble the parts with interference, *heating* or *cooling* may be required.
- Used in:
 - keyed pulley and Shaft,
 - *Car wheels*, etc.
- The interference fits may be:
 - Shrinking fit,
 - Heavy drive fit, and
 - Light drive fit.



❖ The value of minimum interference is $30.25 - 30.30 = -0.05$ mm and maximum interference is $30.15 - 30.40 = -0.25$ mm.

Types of interference fit

Shrink Fit or Heavy Force Fit

- It refers to *maximum negative allowance*.
- In assembly of the hole and the shaft, the hole is *expanded by heating* and then *rapidly cooled* in its position.
- It is used in fitting of *rims* etc.,

Medium Force Fit

- These fits have *medium negative allowance*.
- Considerable *pressure* is required to assemble the *hole* and the *shaft*.
- It is used in *car wheels, armature of dynamos* etc.,

Tight Fit or Force Fit

- One part can be assembled into the other with a *hand hammer* or by *light pressure*.
- A *slight negative allowance* exists between *two mating* parts.
- It gives a *semi permanent fit* and is used on a *keyed pulley* and *shaft, rocker arm*, etc.,

Fit classifications

Types of fit	Maximum allowance	Minimum allowance
Clearance fit	+ve	+ve
Transition fit	+ve	-ve
Interference fit	$\pm$ ve	-ve

Dimensioning of assembled parts

- When assembled parts are dimensioned, the fit is indicated by the **basic size common to both the components**, followed by the hole tolerance symbol first and then by the shaft tolerance symbol
- E.g., $\Phi 25 H7/h6$, etc.,*

H-internal
feature/holes/

h-external
feature/shaft/

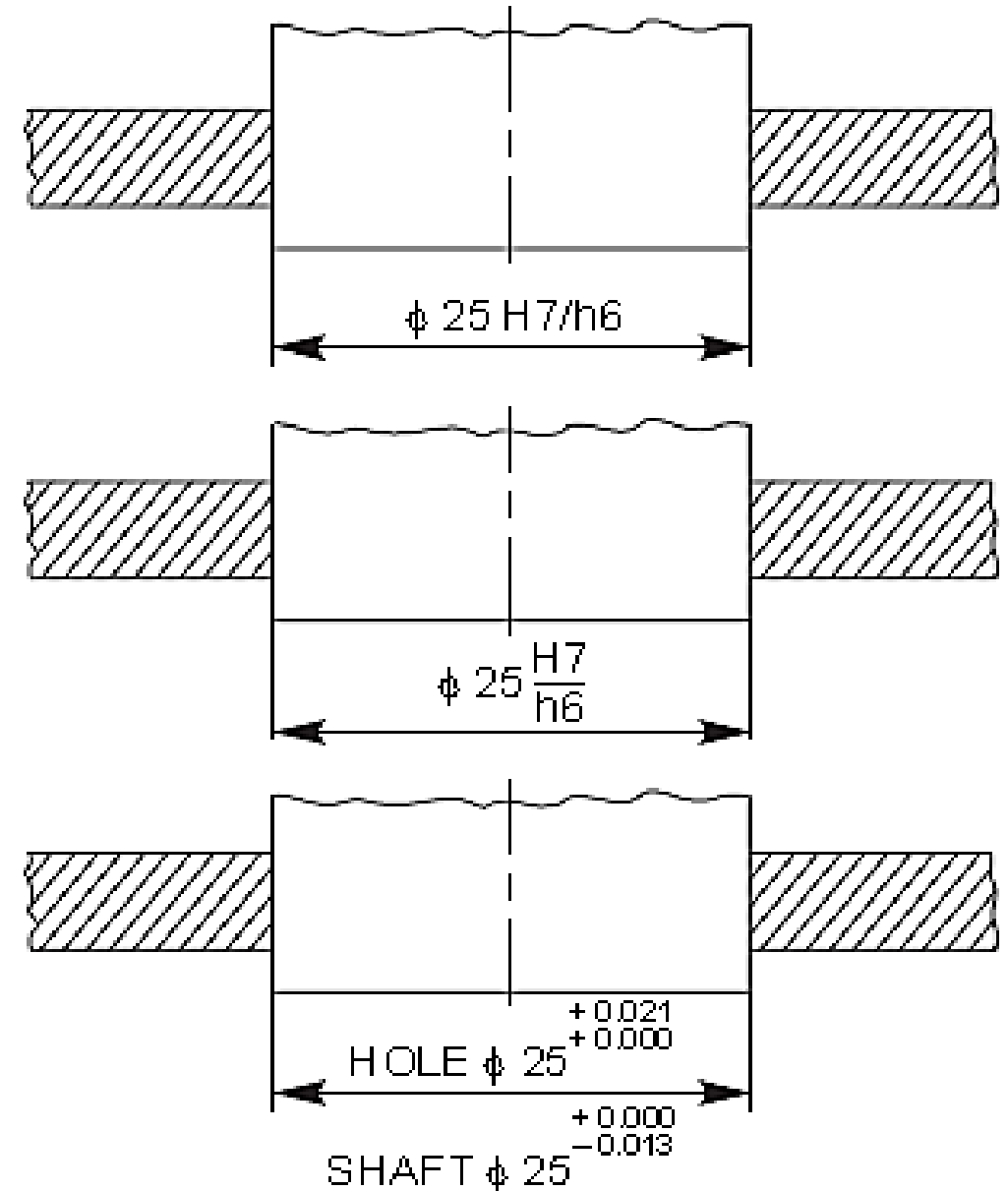
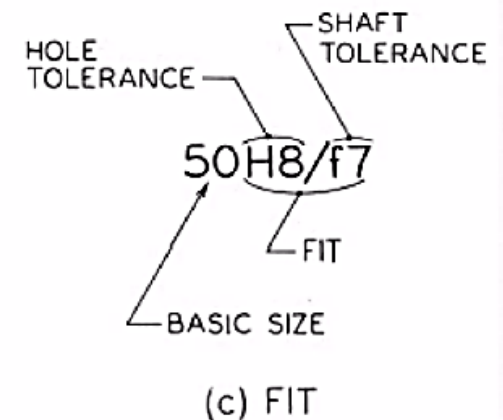
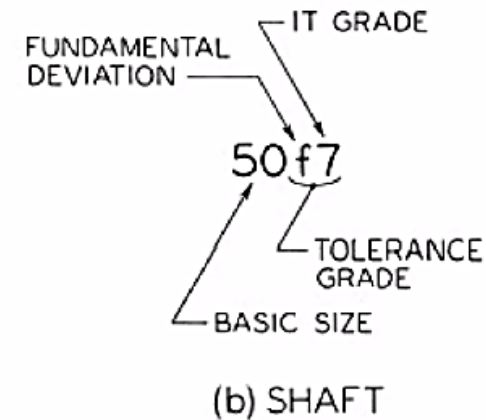
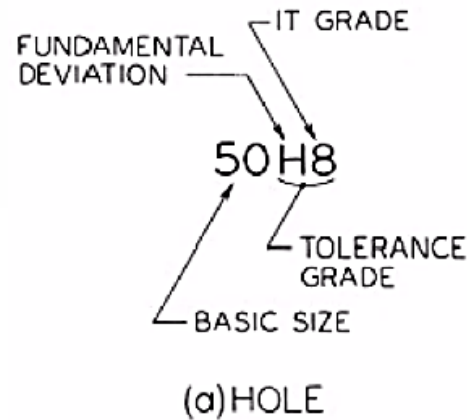
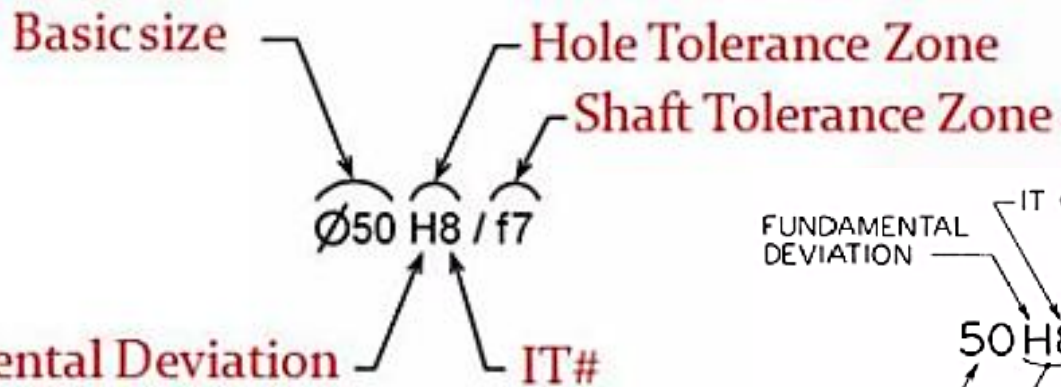


FIG Toleranced dimensioning of assembled parts

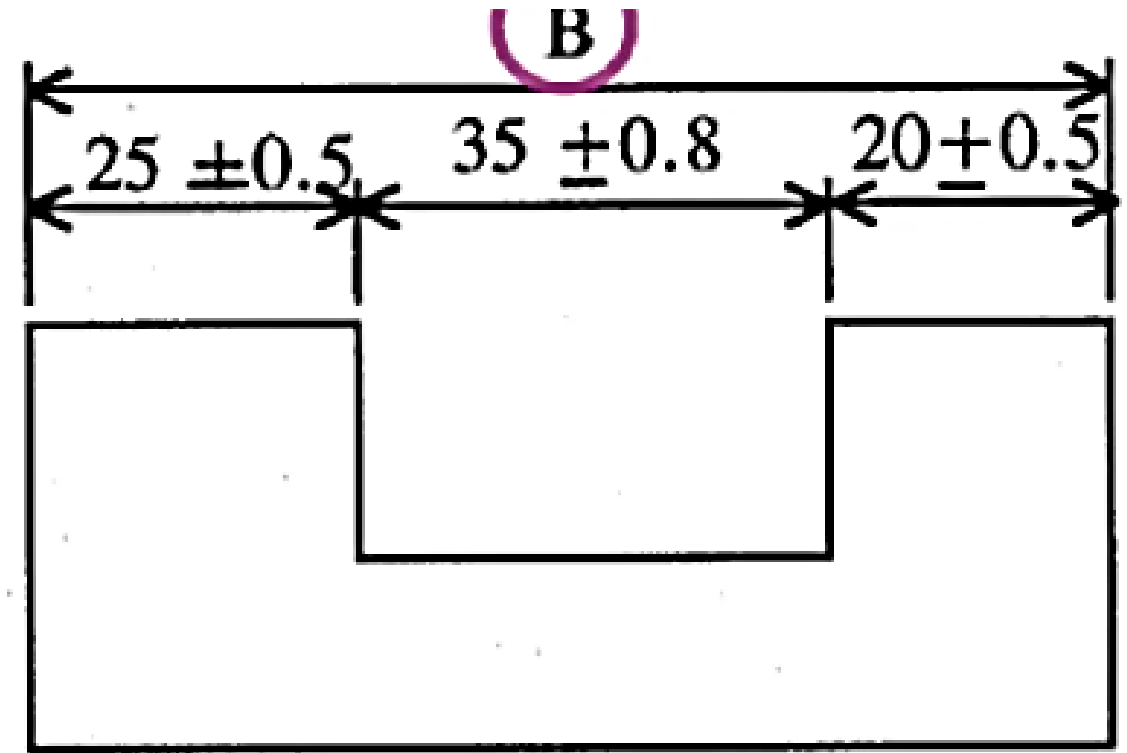
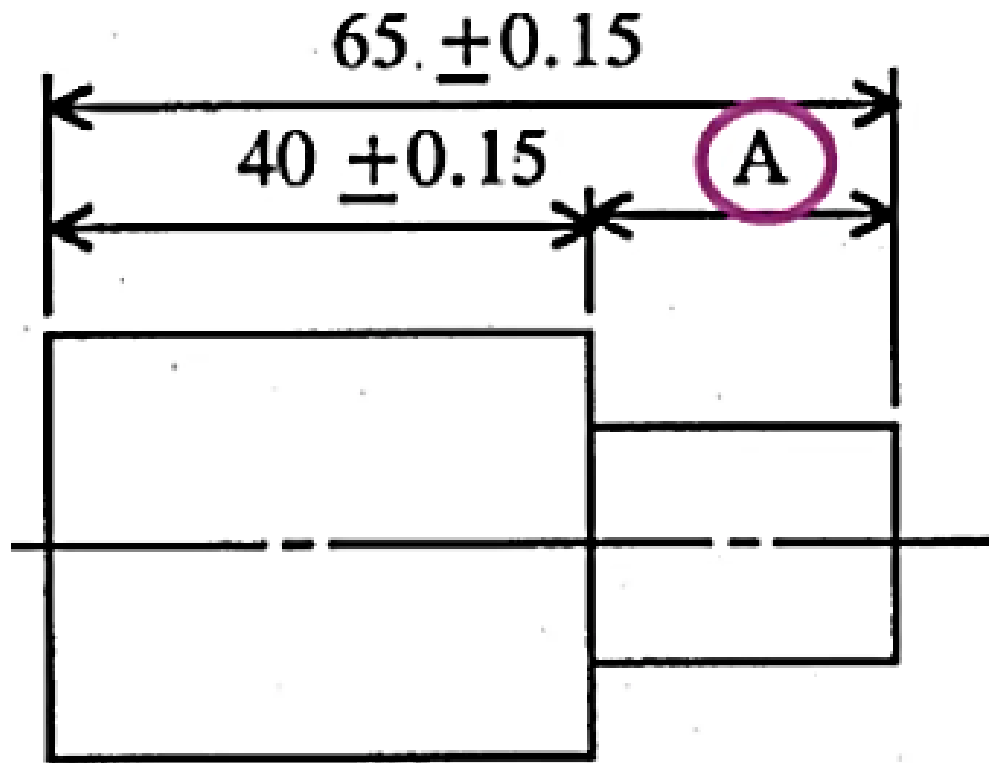
Tolerance Grades

Grade of Tolerance: It is an indication of the level of accuracy.

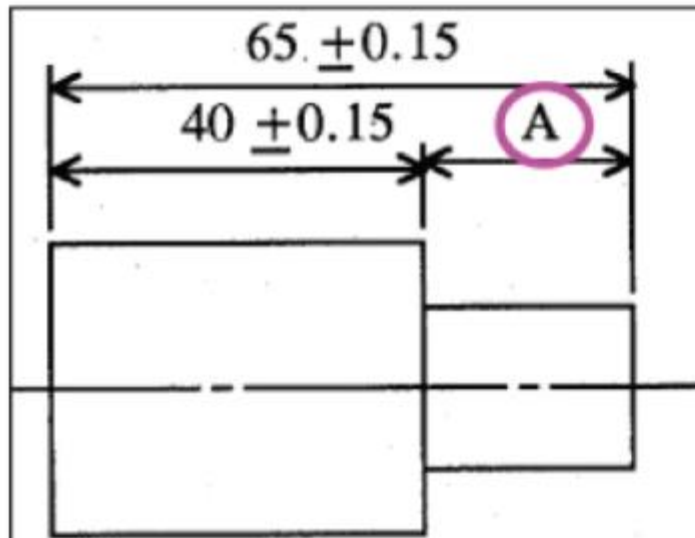
- IT5 to IT10: For *production of gauges, plug gauges, measuring instruments*.
- IT8 to IT11: For fits in *precision engineering* applications.
- IT10 to IT11: For *General Engineering*.
- IT11 to IT16: For *Sheet metal working* or *press working*.
- IT15 to IT16: For processes like *casting, general cutting work*.



Class work



Class work

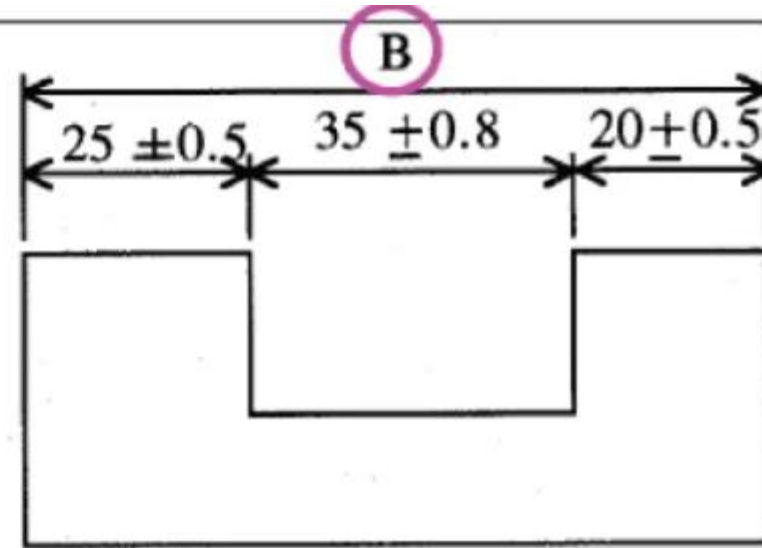


$$A = 65 \pm 0.15 - 40 \pm 0.15$$

$$= 25 \pm 0.3$$

$$A \text{ max.} = 25.3 \text{ mm}$$

$$A \text{ min.} = 24.7 \text{ mm}$$



$$B = 25 \pm 0.5 + 35 \pm 0.8 + 20 \pm 0.5$$

$$= 80 \pm 1.8$$

$$B \text{ max.} = 81.8 \text{ mm}$$

$$B \text{ min.} = 78.2 \text{ mm}$$

Answers

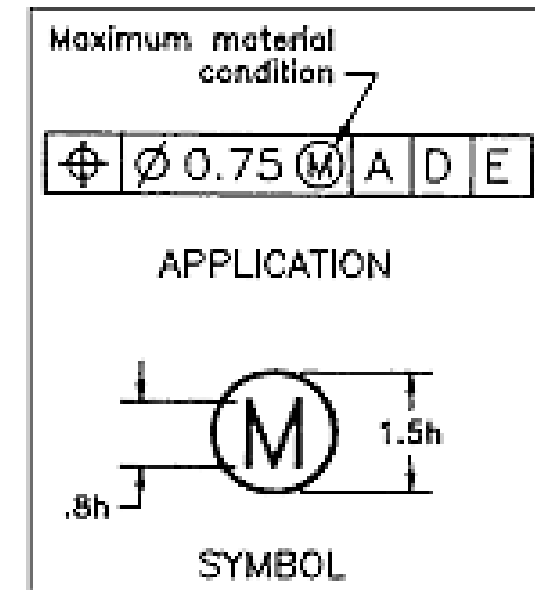
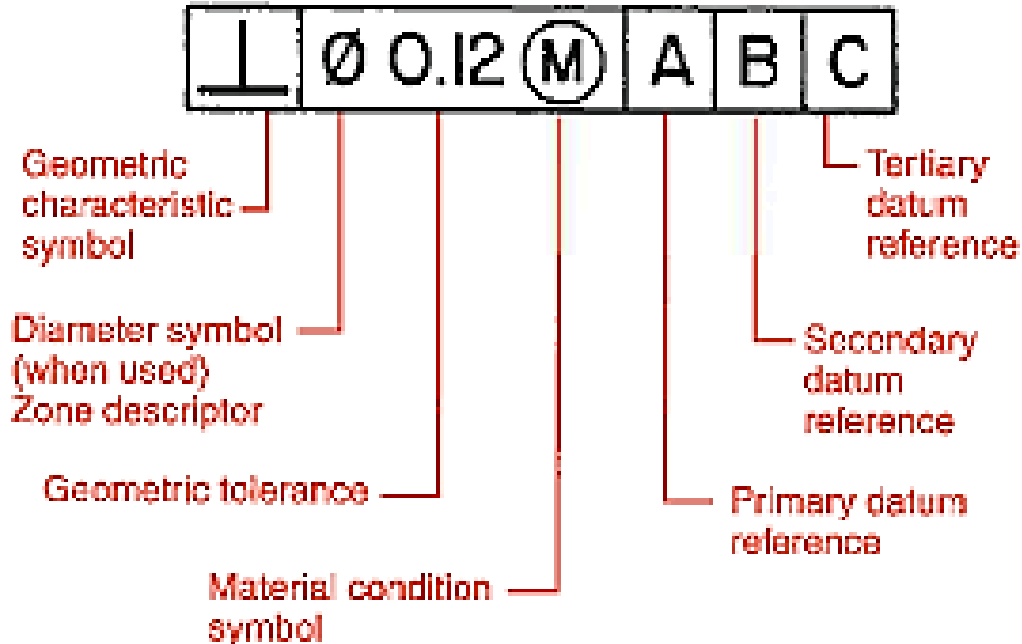
Geometric Tolerance

Indication of Feature Controlled

- A geometric tolerance is prescribed using a feature control frame/Tolerance frame/.
- It has three components:
 - i. The tolerance **symbol**/form symbol/,
 - ii. The tolerance **value**,
 - iii. The **datum labels** for the reference frame.

Tolerance frame

- These indications are written **in rectangular frames** which are **divided into two or three compartments**.
- The compartments are filled from left to right in the order as shown in figure;



Tolerance frame

- The tolerance frame is connected to the feature by a leader line terminating with an arrow head.
- The arrow head of the leader line may be placed
 1. At the **outline of the feature**, to be tolerance(fig 'a') or
 2. On the **extension line** (figure 'b' & 'c') or
 3. On the **axis** (figure d).

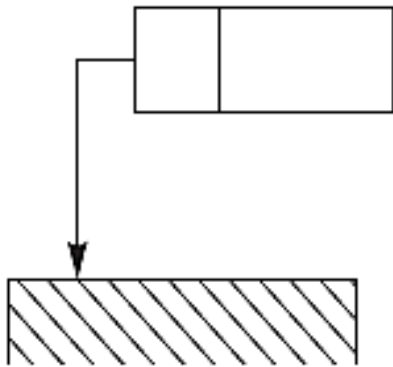


Figure 'a'

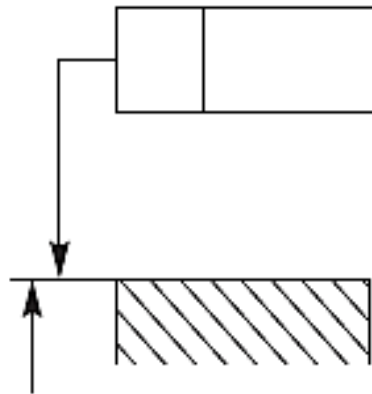


Figure 'b'

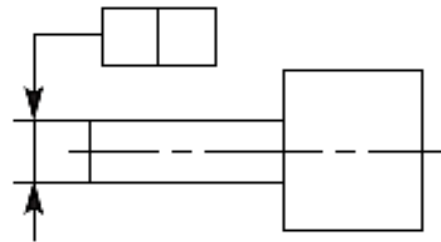


Figure 'c'

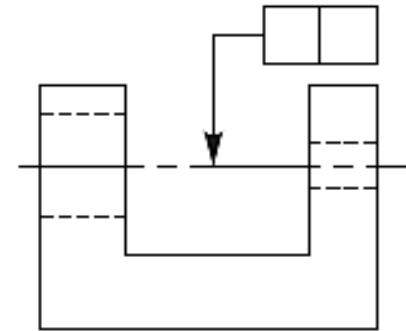


Figure 'd'

Datum feature

- The datum features are connected to the tolerance frame by a leader line terminating with a solid or open triangle.
- The base of the triangle lies on the **outline** of a feature or an **extension line** or on the **axis** as shown in figure 'e' and 'f'.

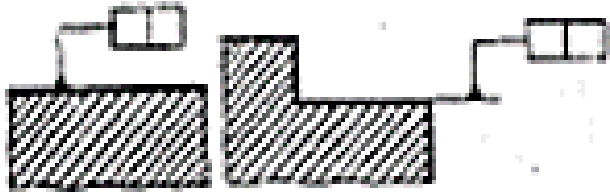


Fig. e

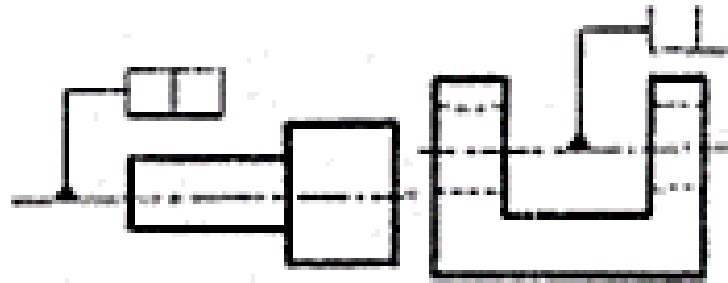
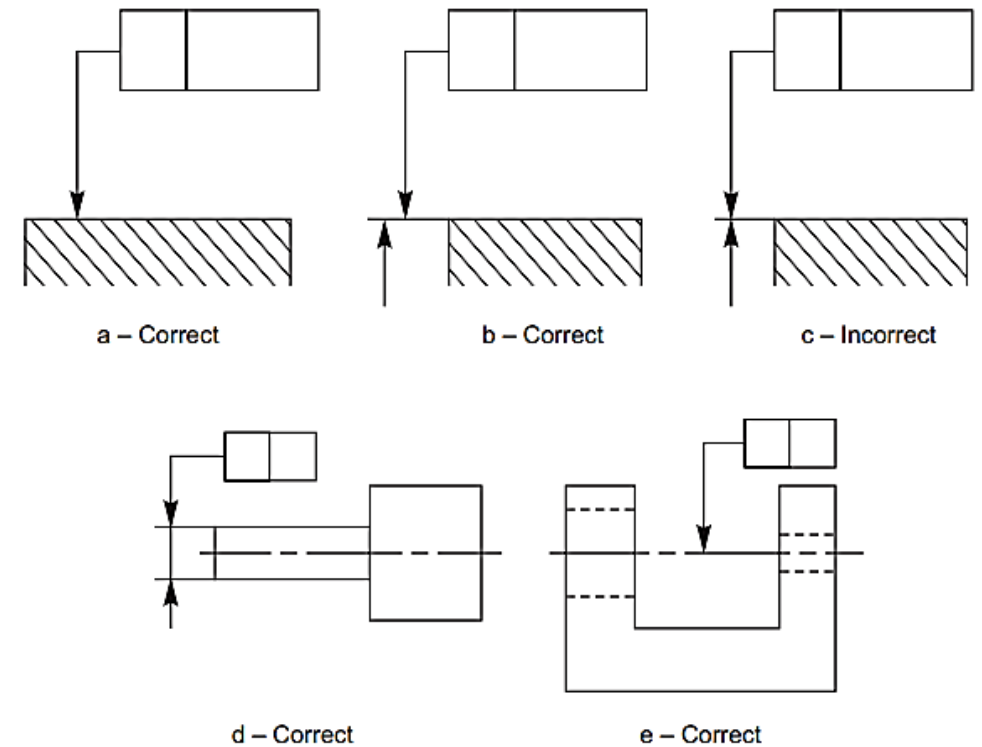
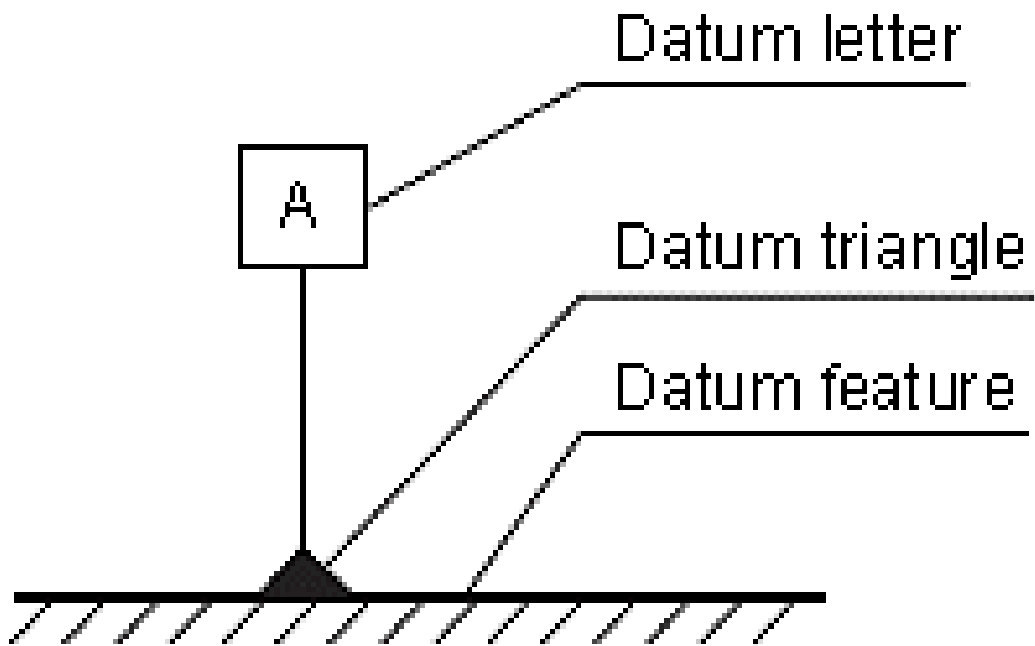


Fig. f

Datum feature

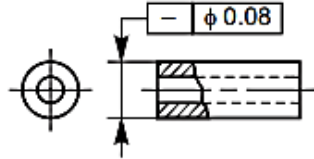
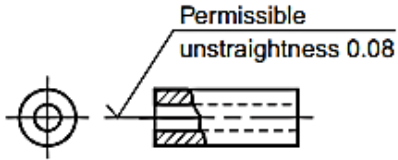
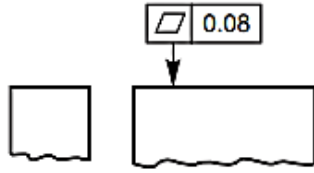
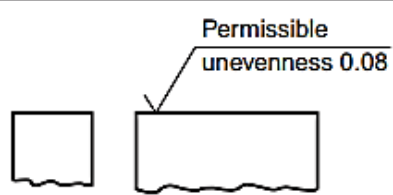
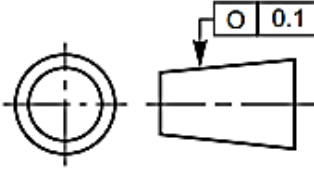
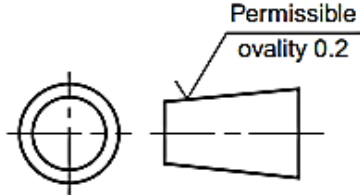
- Sometimes it may not be possible to connect the datum feature with a leader line [Examples: grooves, slots etc...].
- In such cases, a capital letter written in a frame is connected by a leader line terminating with a filled or open triangle whose base is placed on the datum feature as shown in figure.

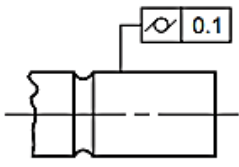
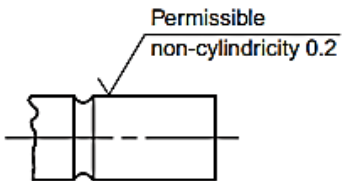
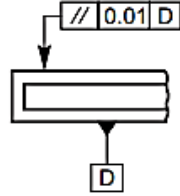
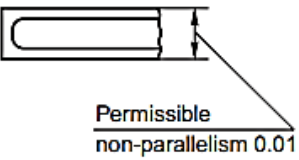
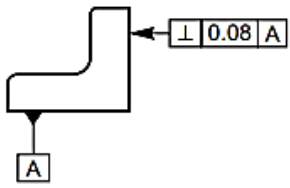
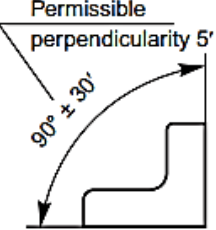


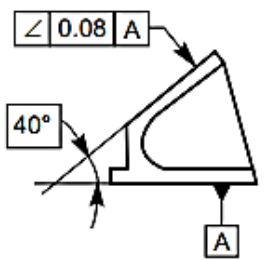
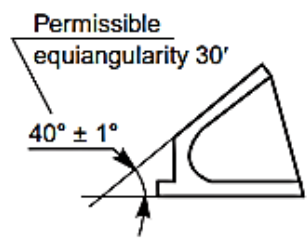
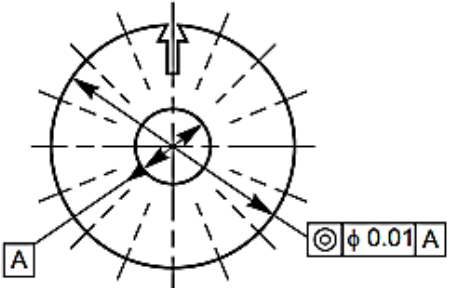
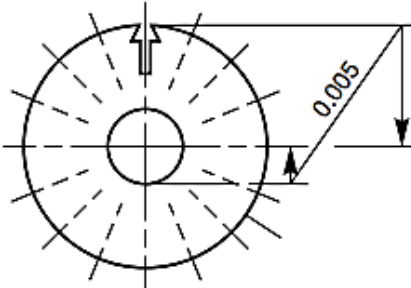
Symbols for geometrical characteristics

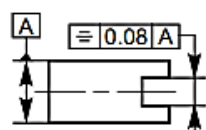
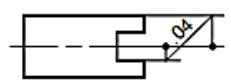
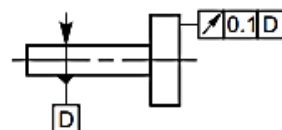
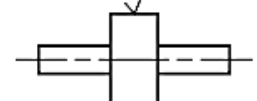
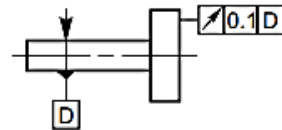
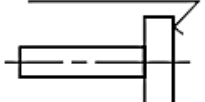
<i>Characteristics to be toleranced</i>		<i>Symbols</i>
Form of single features	Straightness	
	Flatness	
	Circularity (roundness)	
	Cylindricity	
	Profile of any line	
	Profile of any surface	
Orientation of related features	Parallelism	
	Perpendicularity (squareness)	
	Angularity	
Position of related features	Position	
	Concentricity and coaxiality	
	Symmetry	
	Run-out	

Standards followed by industries

As per the standard	As prevalent in industry
1. Straightness tolerance	
	
2. Flatness tolerance	
	
3. Circularity tolerance	
	

4. Cylindricity tolerance	
	
5. Parallelism tolerance	
	
6. Perpendicularity tolerance	
	

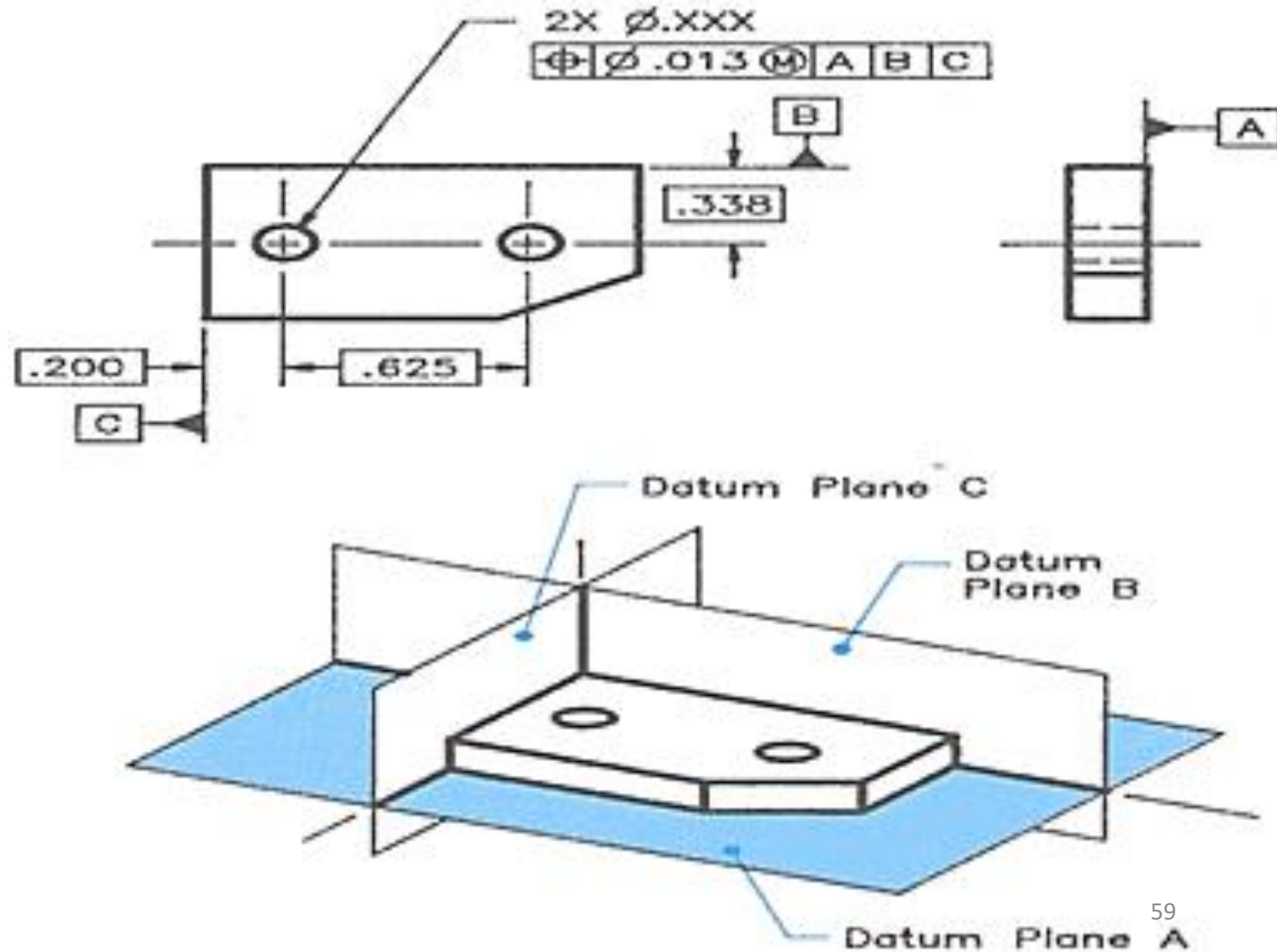
7. Angularity tolerance	
	
8. Concentricity and coaxiality tolerance	
	

9. Symmetry tolerance	
	
10. Radial run-out	
	
11. Axial run-out	
	

Reference Frame

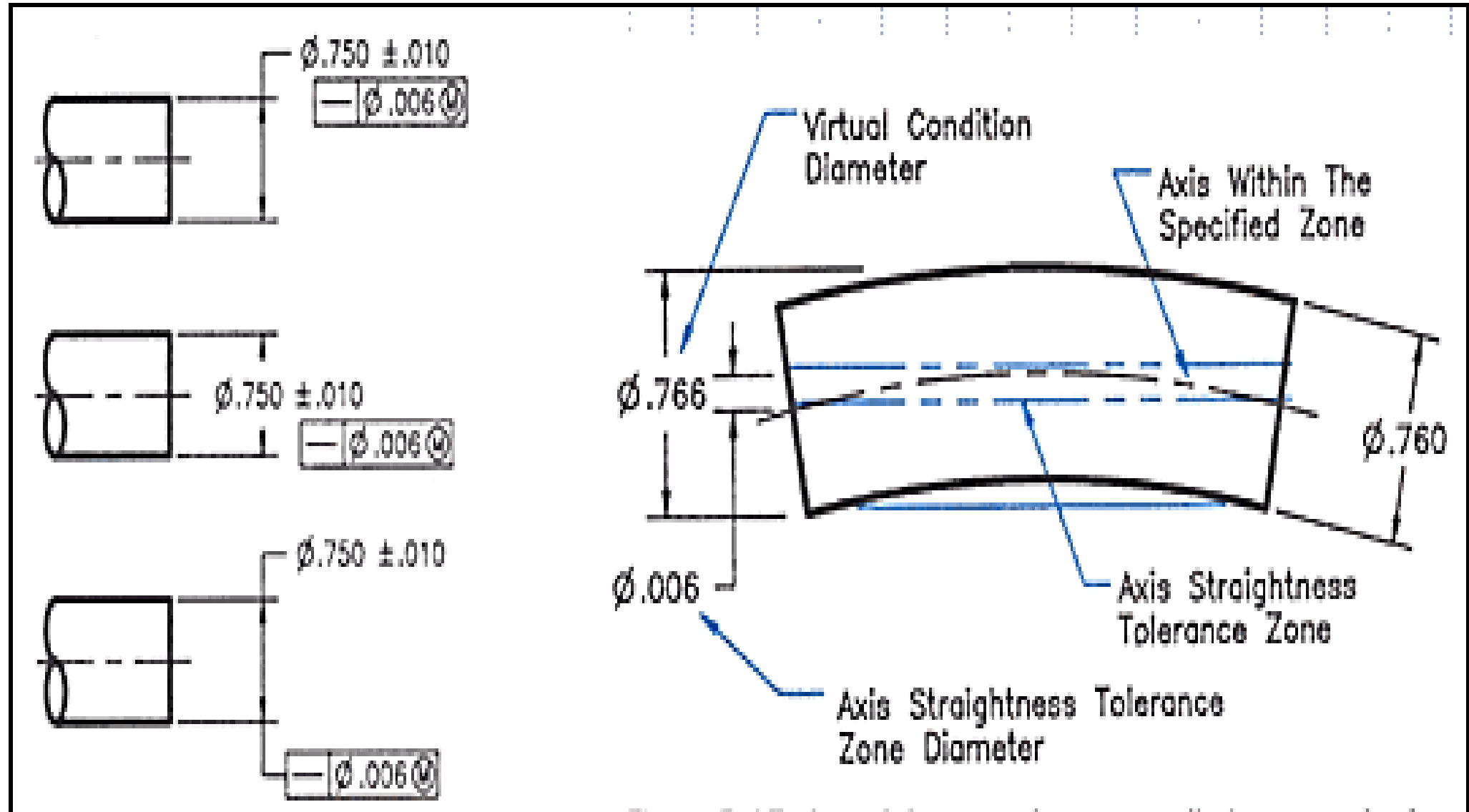
- A reference frame is defined by *three perpendicular* datum planes.

Example 1



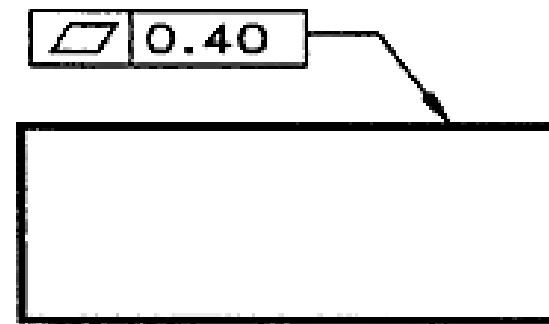
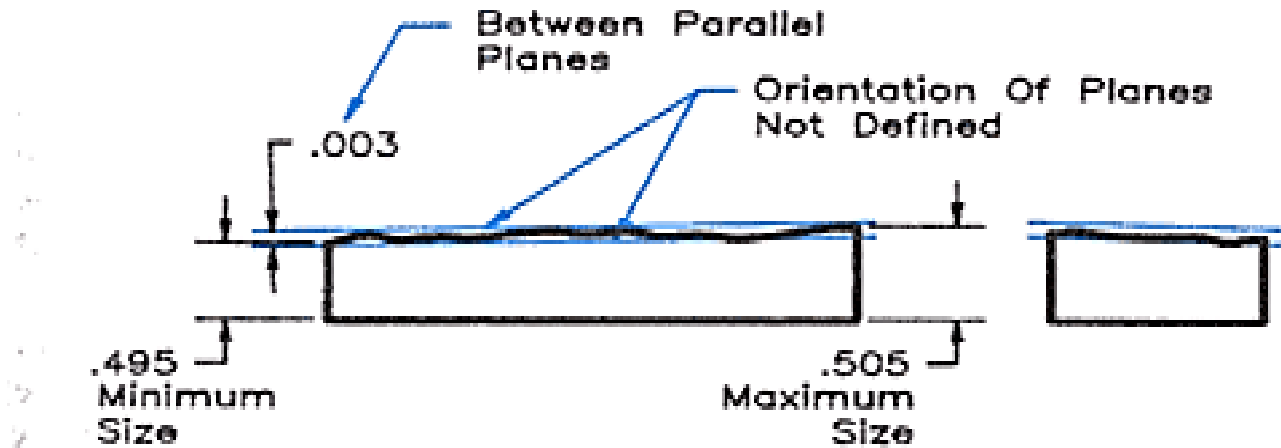
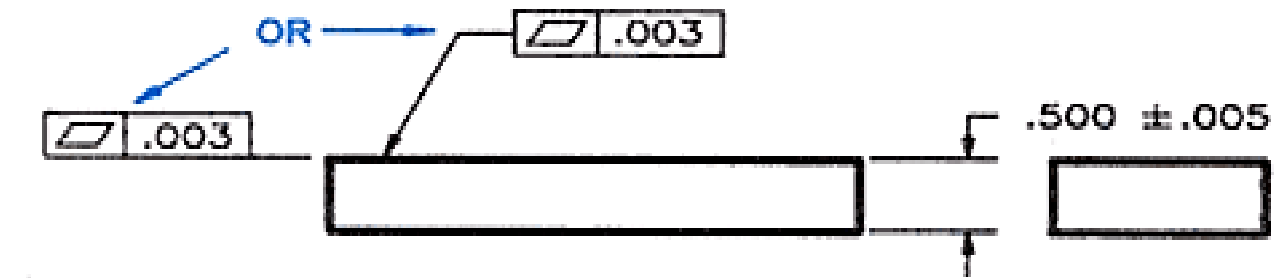
Straightness tolerance

Example 2



Flatness

Example 3



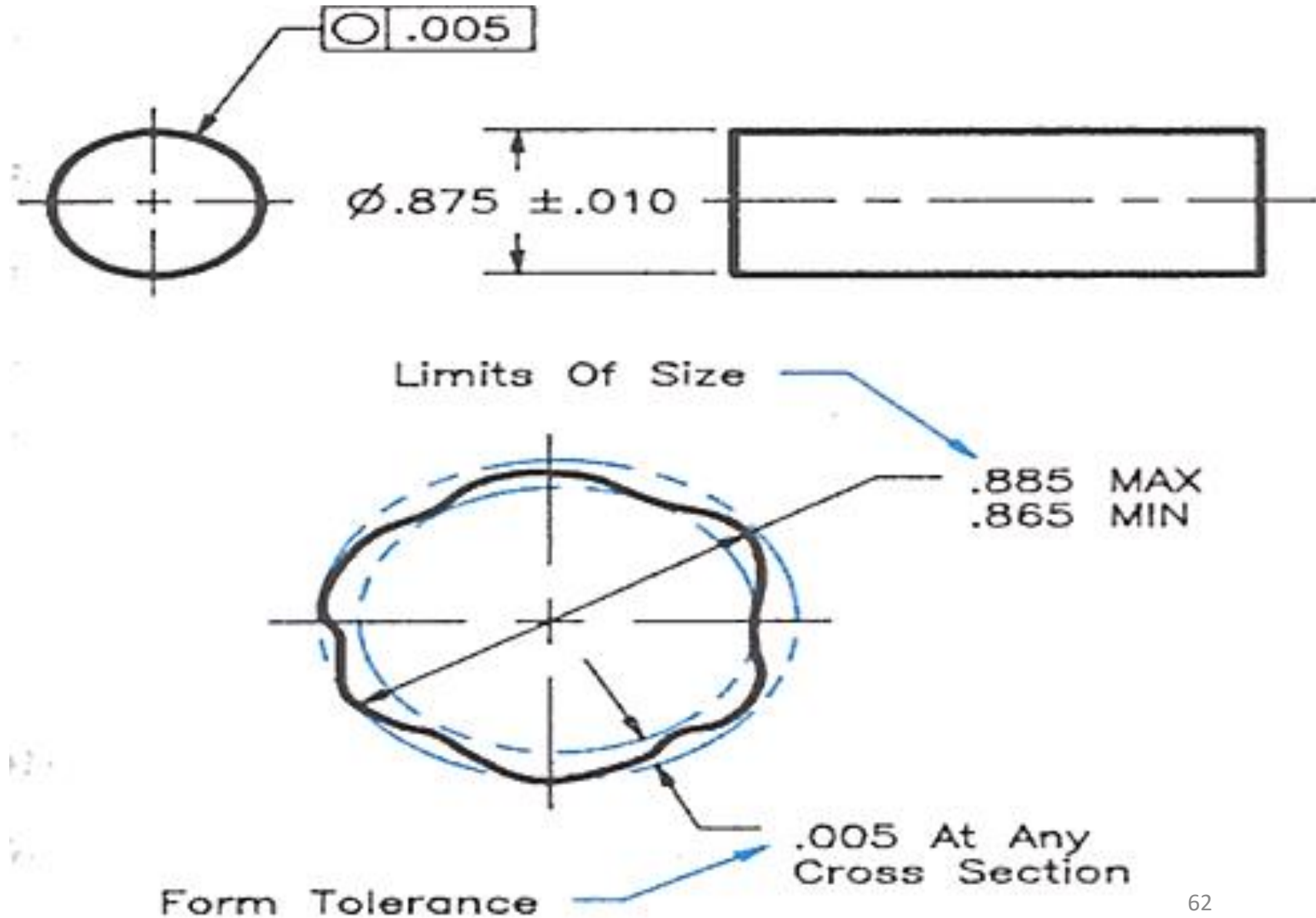
A. DRAWING



B. MEANING

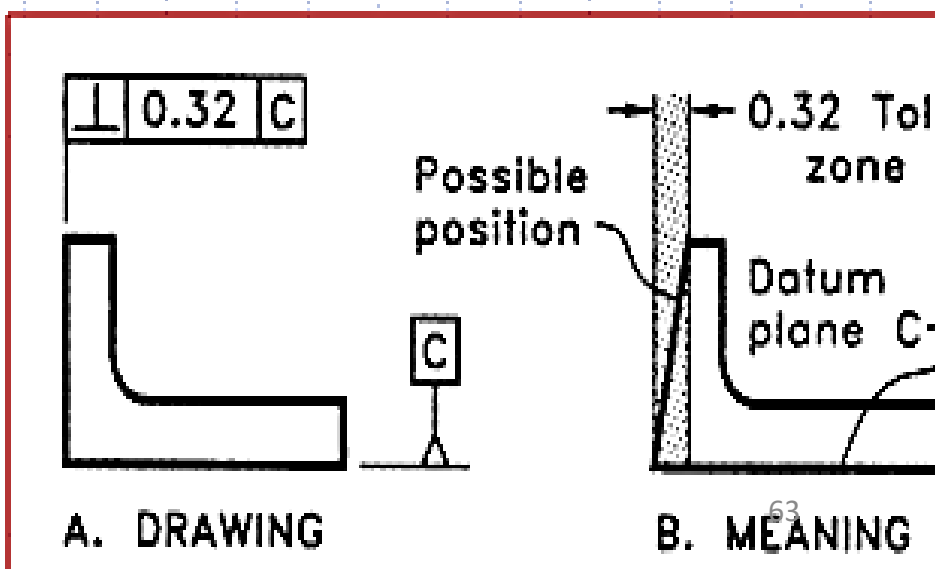
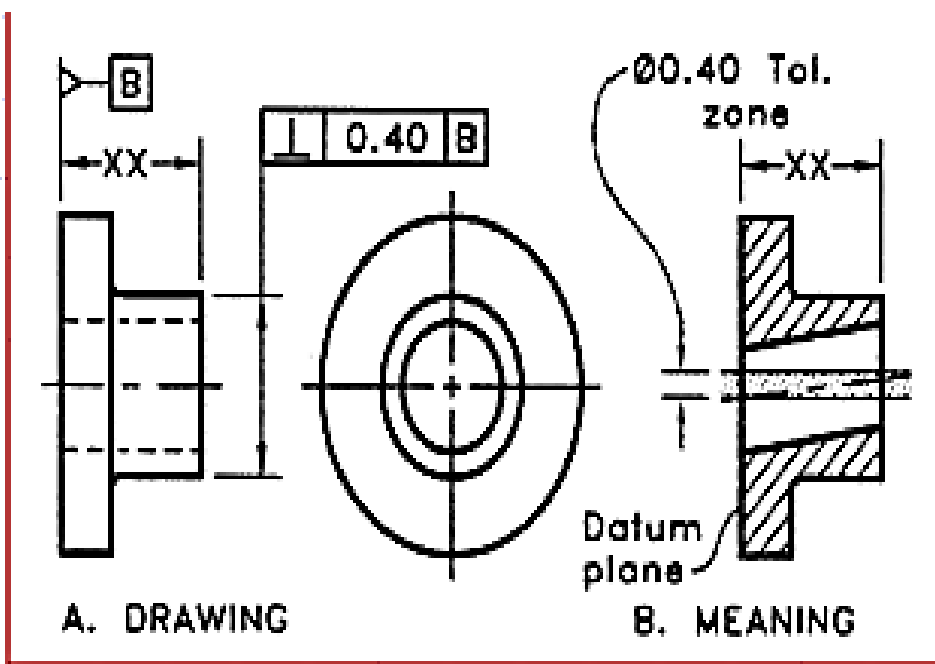
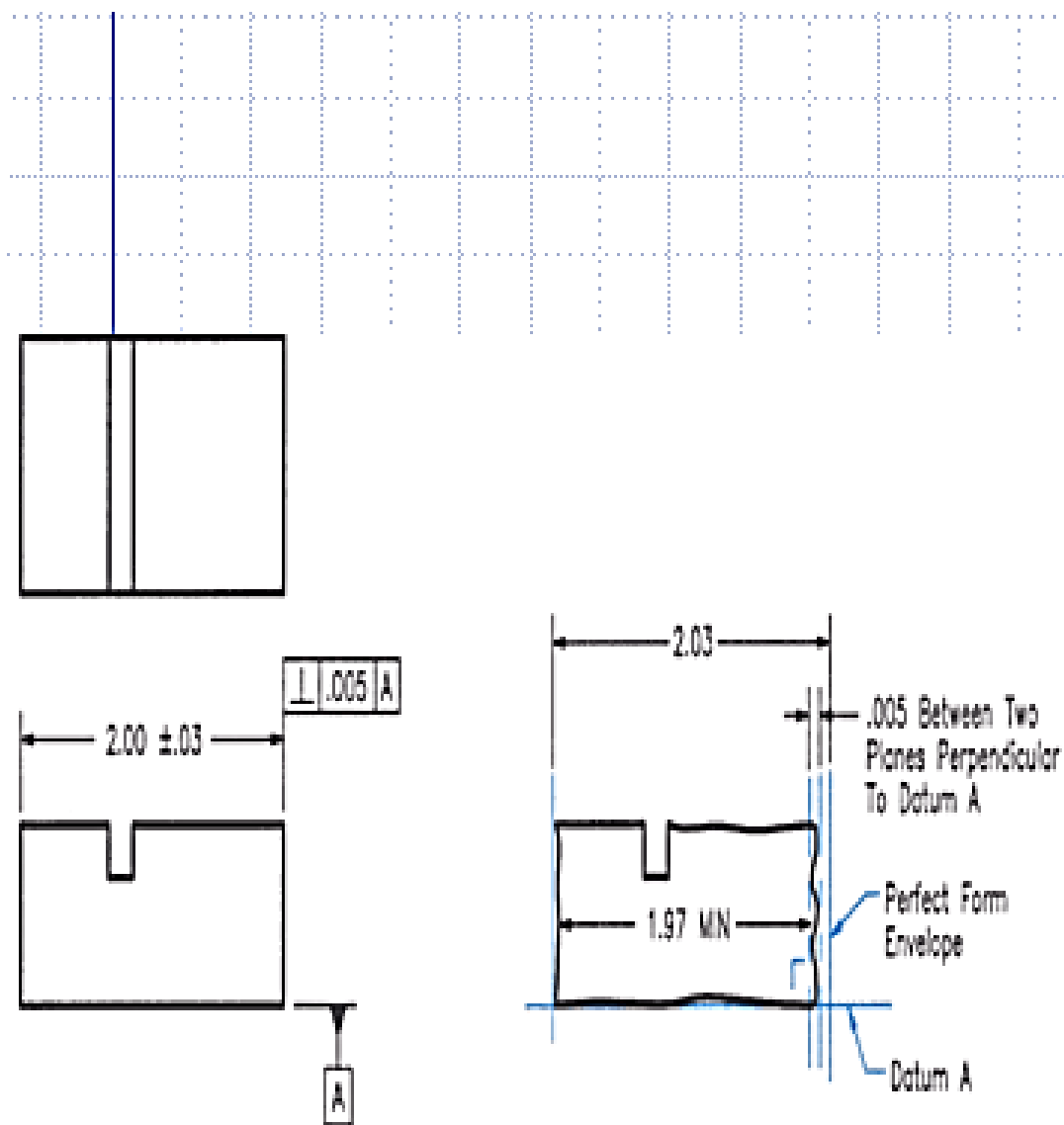
Circularity tolerance

Example 4



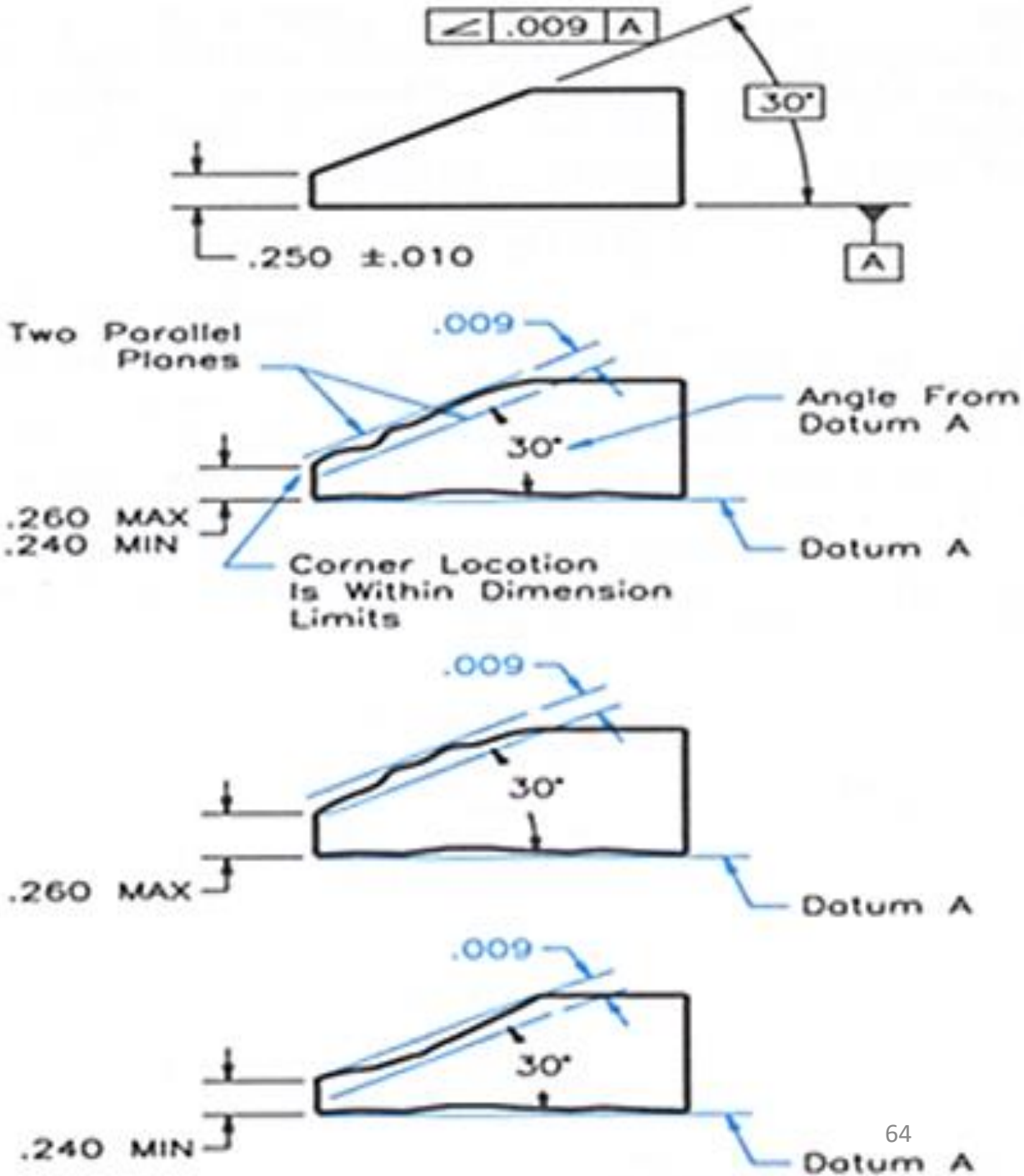
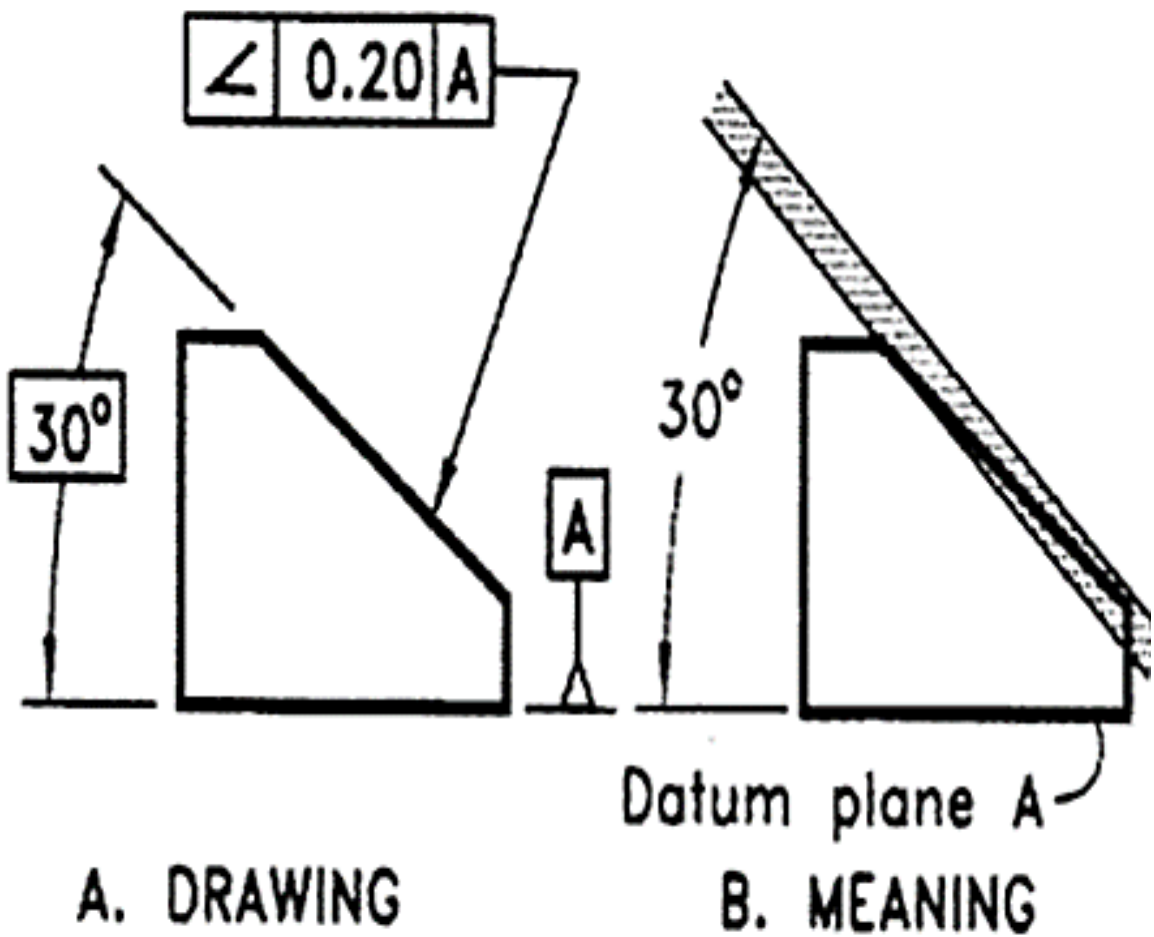
Perpendicularity Tolerance

Example 5



Angular tolerances

Example 6



Thank You